

COMSATS University Islamabad, Pakistan

CUI Virtual Guide

**A project presented to
COMSATS University Islamabad Sahiwal Campus**

**In partial fulfillment
of the requirement for the degree of**

Bachelor of Science in Computer Science (2022-2026)

By

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It is to certify that the final year project of BS (CS) “CUI Virtual Guide” was developed by **Alina Fazal (CIIT/SP22-BCS-024)**, **Alina Javed (CIIT/SP22-BCS-025)** and **Anoosha Gull (CIIT/SP22-BCS-026)** under the supervision of “**DR. MUHAMMAD SHOAIB**” and co supervisor “**MR. MUHAMMAD JAMIL**” and that in their opinion; it is fully adequate, in scope and quality for the degree of Bachelor of Science in Computer Sciences.

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Executive Summary

The CUI Virtual Guide serves as a centralized platform offering essential campus services such as interactive maps, departmental information, faculty directories, academic timetables, announcements, and student support resources. Built using modern technologies including an Android-based frontend developed in Android Studio using Kotlin, with UI/UX prototypes designed in Figma ensuring smooth interaction, accessibility, and a modern mobile experience and a structured backend with database support the system ensures scalability, reliability, and an intuitive user experience.

A key component of the project is the integration of an **AI-based chatbot**, enabling users to quickly access answers to common queries such as course information, building locations, fee-related questions, and general campus procedures. This intelligent assistant reduces the workload on administrative staff and provides instant support, improving overall accessibility and efficiency.

The system also includes an admin panel that enables university staff to manage and update information such as timetables, announcements, maps, and department details. This ensures that all content remains accurate, up-to-date, and easily modifiable. Additional future enhancements may include navigation-based features, real-time notifications, API integration for academic portals, and personalized recommendations for students.

Overall, **CUI Virtual Guide** aims to be a complete digital navigation and information solution that combines modern design, intelligent automation, and robust functionality to enhance the campus experience and support the academic community effectively.

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Acknowledgement

All praise is to Almighty Allah who bestowed upon us a minute portion of His boundless knowledge by virtue of which we were able to accomplish this challenging task.

We are greatly indebted to our project supervisor “Dr. Muhammad Shoaib” and our co-Supervisor “Mr. Muhammad Jamil”. Without their personal supervision, advice and valuable guidance, completion of this project would have been doubtful. We are deeply indebted to them for their encouragement and continual help during this work.

And we are also thankful to our parents and family who have been a constant source of encouragement for us and brought us the values of honesty & hard work.

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Abbreviations

CUI	Comsats University Islamabad
VG	Virtual Guide
UI	User Interface
UX	User Experience
API	Application Programming Interface
QR	Quick Response Code
TTS	Text-to-Speech
XML	Extensible Markup Language
JSON	JavaScript Object Notation
AI	Artificial Intelligence
CRUD	Create, Read, Update, Delete
MVC	Model View Controller
SDK	Software Development Kit
URL	Uniform Resource Locator

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Chapter 1

Introduction

1. Introduction

This chapter provides a comprehensive introduction to the final year project titled “CUI Virtual Guide”, which is an Android-based mobile application developed to assist students, faculty, and visitors of COMSATS University Islamabad by offering centralized access to essential academic and campus-related information. The chapter outlines the background, objectives, methodology, academic relevance, reviewed literature, challenges faced during development, and the justification behind the selected tools and technologies.

CUI Virtual Guide is designed to be a user-friendly, visually appealing, and functionally robust digital assistance platform that reduces dependency on physical guidance desks, printed schedules, and manual inquiry processes. The idea originated from the growing need for a modern, accessible solution that could simplify navigation within the campus, streamline information retrieval, and improve communication between the institution and its users.

The primary goal of this project is to offer a unified mobile application where users can easily access services such as departmental details, faculty directories, academic timetables, campus maps, general announcements through real-time web scrapping, and student support resources. In addition, the system integrates an AI-powered chatbot capable of responding to frequently asked questions and guiding users in real time enhancing productivity and reducing administrative workload.

Prototypes and user interface designs were created using Figma, ensuring a clean, intuitive, and consistent layout. The mobile application was developed in Android Studio using Kotlin, selected for its modern programming features, strong Android support, and efficient performance. Together, these tools enable the system to deliver seamless navigation, improved usability, and an enhanced overall user experience.

1.1. Project and Brief

CUI Virtual Guide is an Android-based mobile application developed to assist students, faculty, and visitors of COMSATS University Islamabad by providing centralized access to essential campus-related information. This project aims to overcome the challenges associated with manual inquiries, outdated printed schedules, and difficulty in navigating university services by offering a unified, smart, and user-friendly digital solution.

The application includes major features such as academic timetables, faculty directories, departmental details, campus navigation assistance, general announcements, and a built-in AI-powered chatbot for real-time support. The chatbot enables users to receive instant responses to frequently asked questions, reducing dependency on physical help desks and improving the overall communication experience within the campus environment.

The technologies used in this project include Android Studio for application development, Kotlin as the primary programming language due to its modern syntax and strong Android support, and Figma for creating the UI/UX prototypes to ensure a clean and interactive design. The application structure follows a well-organized modular architecture, allowing for scalable development and future feature extensions.

The development process follows an Agile Software Development Methodology (SDLC) model, enabling the system to be built and tested in small functional modules. This approach ensures continuous improvements, early detection of issues, and smooth integration of components such as the timetable module, information directory, and AI chatbot.

This report provides an in-depth explanation of the system's architecture, design methodology, development workflow, and the implementation of intelligent assistance features that collectively enhance user experience and accessibility across the COMSATS University campus.

1.2. Relevance to Course Modules

This project demonstrates the practical application of several theoretical and technical modules studied throughout the academic program. The key subjects that directly contributed to the development of the **CUI Virtual Guide** include:

- **Mobile Application Development:** Applied to build the Android application using **Android Studio** and **Kotlin**, implementing concepts like activity lifecycle, UI layouts, adapters, and data handling.
- **Software Engineering:** The project follows the **Agile Software Development Methodology**, allowing iterative progress, continuous feedback, flexible requirement handling, and faster delivery of functional modules such as the timetable, directory, and chatbot.

- **Database Systems:** Concepts of relational data modelling were used to structure and manage campus information, including faculty details, department data, announcements, and schedules using Firebase as the backend for storage and real-time updates.
- **Human-Computer Interaction (HCI):** HCI principles guided the creation of clean, user-focused UI designs in Figma, ensuring accessibility, simple navigation, and a visually appealing interface.
- **Artificial Intelligence:** Fundamental AI concepts were applied to integrate an AI-based chatbot that assists users by answering frequently asked questions and providing campus guidance.
- **Networking & Communication:** Used to handle real-time data retrieval, interaction between app modules, and communication with backend services.

1.3. Project Background

The CUI Virtual Guide project originated from the increasing need for digital assistance within university environments, where students, faculty, and visitors often struggle to access accurate and centralized campus information. Traditional methods such as physical notice boards, manual inquiries, and scattered information sources are time-consuming, unreliable, and insufficient in a modern academic ecosystem. New students, in particular, face challenges navigating the campus, locating departments, finding faculty offices, and understanding university procedures.

To address these issues, the idea of a mobile-based virtual guide was proposed. The goal is to create a single, unified digital platform that provides essential information in an interactive, efficient, and user-friendly manner. With the increasing reliance on smartphones, an Android-based solution becomes the most accessible and practical approach for students and visitors.

The project also takes inspiration from the growing trend of smart campuses and digital academic support systems around the world. However, many institutions including COMSATS University Islamabad (CUI) still lack a comprehensive mobile guide that brings together navigation features, academic information, chatbot support, event updates, and communication tools in one place.

This system aims to bridge that gap by offering a modern, scalable virtual guide that simplifies campus navigation through a custom-built Figma map and coordinate-based routing, eliminates repeated manual inquiries through an AI chatbot, and centralizes important

university information for easier access. Designed with future expansion in mind, the CUI Virtual Guide serves as a foundation for more advanced smart-campus functionalities that can be added later.

1.4. Literature Review

To better understand the current state of digital guide systems and campus navigation solutions, a review of existing platforms was conducted. Several existing systems were analyzed in terms of interface usability, information accessibility, navigation features, and interactive capabilities.

For instance, UniMap focuses primarily on providing static campus maps but lacks interactive guidance or detailed information about departments, faculty, and events. Its interface does not offer step-by-step navigation, which makes it difficult for new students or visitors to efficiently find locations on campus. CampusGuide.pk, on the other hand, provides textual information about university departments but lacks real-time updates, multimedia assistance, and QR code integration, limiting user engagement. Lastly, SmartCampus, while effective for announcements and notifications, does not provide location-based guidance or voice-assisted interaction, making it unsuitable for users seeking practical, on-the-spot assistance.

These observations highlight a major gap in existing solutions especially for systems that offer interactive, real-time, multimedia-enabled guidance for students, faculty, and visitors. The **CUI Virtual Guide** addresses this gap by providing a comprehensive navigation system, an intuitive interface, voice assistant features, QR code integration, and quick access to campus-related information.

- **UniMap** primarily focuses on static campus layouts. It lacks interactive guidance or step-by-step navigation, which limits usability for new visitors.
- **CampusGuide.pk** provides basic information about departments and faculty but does not include real-time interaction, QR code support, or AI-assisted guidance. This reduces efficiency for users needing immediate help.
- **SmartCampus** offers notifications and announcements but does not provide a full-featured guide or interactive navigation. It is not designed for first-time visitors or users seeking hands-on assistance within campus.

The review shows that while several systems provide partial campus support, none combine navigation, academic information, and AI-powered assistance in a single, user-friendly

platform. The **CUI Virtual Guide** fills this gap by integrating interactive campus navigation, departmental and faculty details, timetables, announcements, and an AI chatbot to assist users in real time.

1.5. Analysis from Literature Review

Based on the findings from the literature review, several improvements and features were incorporated into the CUI Virtual Guide to provide a superior user experience for students, faculty, and visitors:

- The platform offers comprehensive campus information, including departmental details, faculty directories, academic timetables, announcements, and student support resources.
- Interactive navigation and a custom-designed campus map were implemented, enabling users to find locations efficiently using manually mapped coordinates.
- The design is responsive and device-compatible, allowing smooth access across different Android devices.
- An AI-powered chatbot was integrated to provide real-time guidance, answer frequently asked questions, and assist users in navigating the campus.
- Features such as a Contact Us page, feedback submission module, and event information section enhance communication between users and the university administration.

These enhancements ensure that the CUI Virtual Guide not only addresses the gaps identified in existing campus guide systems but also delivers a modern, interactive, and user-friendly solution that centralizes essential campus services and simplifies navigation within COMSATS University Islamabad.

1.6. Methodology and Software Lifecycle for This Project

The Agile Software Development Methodology was selected as the most suitable approach for the **CUI Virtual Guide** kiosk system. Agile emphasizes iterative development, continuous feedback, and adaptability, which is ideal for projects requiring user-centred design and interactive functionality. The development process was divided into small, functional sprints, with each sprint focusing on specific modules such as the user interface and layout design for the touchscreen kiosk, the creation of the interactive campus map and navigation system, the administrative login and authentication module for managing events, announcements, and

feedback, as well as the implementation of departmental details, faculty directories, academic timetables, event management, and an AI-powered chatbot to assist users with frequently asked questions.

Each sprint involved design, development, testing, and review stages, allowing early detection of issues and continuous improvement. This iterative approach ensured that each module could be refined based on feedback before full integration into the kiosk system. The Agile methodology provided several advantages for this project. Continuous feedback from potential users helped ensure that the kiosk interface met the needs of students, faculty, and visitors. Iterative development enabled the progressive delivery of functional modules, improving usability and interaction. The flexibility of Agile allowed new features or adjustments to be incorporated without affecting the overall system, while integrated testing throughout the development lifecycle enhanced the reliability of interconnected modules such as campus navigation, information retrieval, and chatbot guidance.

Figure 1.1 shows how the Agile methodology was applied to develop the CUI Virtual Guide kiosk system was developed in a structured yet flexible manner, resulting in an interactive, user-friendly, and reliable on-campus guidance solution for COMSATS University Islamabad.

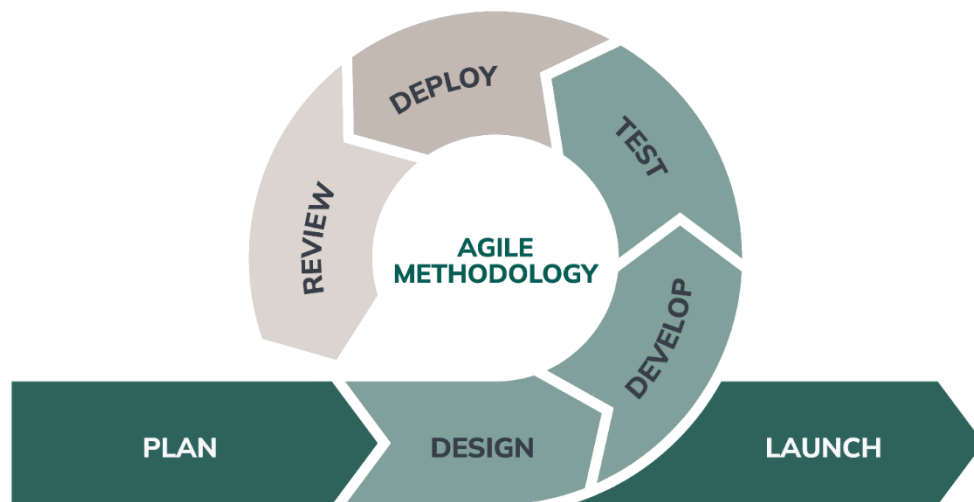


Figure 1.1 Agile Methodology Diagram

1.7. Rational behind Selected Methodology

The rationale for selecting the Agile Software Development Methodology lies in its ability to accommodate evolving project requirements and iterative improvements, which are essential for developing an interactive kiosk system like the CUI Virtual Guide. During the development

process, feedback from potential users, including students, faculty, and visitors, was crucial for refining the interface, navigation, and AI-powered guidance features. Agile provided the flexibility to incorporate such feedback continuously, ensuring that each module, from campus navigation and departmental directories to event management and chatbot functionality, could be improved incrementally without disrupting the overall workflow. Moreover, Agile allowed the development team to divide the system into manageable sprints, focusing on individual modules while testing and refining them before full integration. This approach made it easier to handle interdependencies between modules, such as the interactive map, information retrieval system, and administrative controls. Had a more rigid methodology like Waterfall been chosen, mid-project adjustments such as updating the kiosk interface, integrating new features, or refining the AI chatbot would have been difficult and time-consuming. Agile's iterative nature facilitated smoother testing, improved debugging, and ensured that the system could be evaluated progressively, aligning perfectly with the goals of developing a user-friendly, reliable, and interactive kiosk-based guidance solution for COMSATS University Islamabad.

Chapter 2

Problem Definition

2. Problem Definition

This chapter delves into the specific problems that the **CUI Virtual Guide** addresses, outlines the deliverables and developmental prerequisites of the kiosk system, and examines the existing offline processes that the new platform is designed to replace. It also presents a comprehensive requirements analysis covering both functional and non-functional requirements along with detailed use case descriptions to guide the design and implementation of the system.

2.1. Problem Statement

Students, faculty, and visitors at COMSATS University Islamabad often face difficulty in locating specific classrooms, departments, or blocks, especially if they are new to the campus. Currently, users rely on physical notice boards, printed maps, or manual assistance from staff, which can be time-consuming, confusing, and inefficient. Existing solutions like static maps or textual guides do not provide interactive guidance, real-time navigation, or AI-assisted support, making it challenging for users to quickly find their desired locations.

The primary goal of this project is to develop a kiosk-based CUI Virtual Guide that helps users navigate the campus efficiently. The system will provide interactive maps, step-by-step directions, departmental and faculty information, event updates, and an AI-powered chatbot to guide users in real time. By centralizing campus information and providing an intuitive navigation interface, the kiosk aims to reduce confusion, save time, and enhance the overall campus experience for all users.

2.2. Deliverable and Development Requirements

The following items constitute the key deliverables of the CUI Virtual Guide kiosk system. These outputs ensure that the system meets functional, non-functional, and user requirements. The Agile methodology was chosen for this project because it allows flexibility, iterative development, and early user feedback. Since the kiosk system includes several interdependent modules such as campus navigation, departmental directories, and AI guidance, Agile enabled each module to be developed, tested, and refined progressively, while accommodating new features without redesigning the entire system.

The success of this project hinges on the delivery of the following key components:

1. **Interactive Kiosk Interface:** A touchscreen interface that allows users to easily access campus maps, departmental details, faculty directories, announcements, and events.

2. **Custom Campus Map and Navigation System:** An interactive, coordinate-based campus map that guides users to classrooms, departments, and blocks without relying on external APIs or GPS.
3. **AI-Powered Chatbot:** An integrated chatbot that assists users by answering frequently asked questions and providing real-time guidance within the campus.
4. **Administrative Module:** Secure admin login allowing authorized personnel to manage events, announcements, feedback, and update campus information in the backend database.
5. **Event and Announcement Management:** A system to display real-time updates on campus events, notices, and important announcements for students, faculty, and visitors.
6. **Support and Feedback Module:** A feedback mechanism for users to submit suggestions or report issues, which can be monitored and managed through the admin module.

The development process follows the Agile methodology, allowing each module to be designed, implemented, and tested in isolation before integration. Early iterations focused on core functionality such as interactive navigation and map-based guidance, while later sprints incorporated the AI chatbot, event management, and feedback modules. This iterative approach ensures a reliable, user-friendly, and scalable kiosk system that meets the needs of all campus users.

2.3. Deliverables

- Fully functional CUI Virtual Guide kiosk system
- Interactive touchscreen interface for easy access to campus information
- Custom campus map with navigation to locate classrooms, departments, and blocks
- AI-powered chatbot for real-time guidance and frequently asked questions
- Admin module for managing events, announcements, feedback, and campus information
- Event and announcement display system for up-to-date campus updates
- Feedback submission system to collect user suggestions and issues

2.4. Technical Development Requirements

The following table, Table 2.1, presents the technologies and tools used in the development of the CUI Virtual Guide kiosk system. This style of table numbering and captioning will be followed consistently throughout the report.

Table 2.1: Technologies and Tools Used in the Development of CUI Virtual Guide

Component	Technology/Tool Used
Frontend / User Interface	Figma (prototyping and UI design), touchscreen interface design
Backend	Firebase (real-time database and data management)
Database	Firebase Realtime Database
Development Platform	Android Studio (using Android-based kiosk)
AI & Chatbot Integration	Custom AI chatbot framework
Design Tools	Figma (UI/UX, campus maps, and interactive navigation)

2.5. Current System

The current system at COMSATS University Islamabad is entirely offline and manually managed. Students, faculty, and visitors rely on physical notice boards, printed schedules, static maps, and staff assistance to access campus information, locate classrooms, or find departments and blocks. There is no centralized digital platform to provide real-time guidance or interactive navigation. Information regarding events, announcements, and faculty directories is scattered, making it difficult for users to obtain accurate and timely updates. This manual approach leads to inefficiencies, confusion, and limited accessibility, especially for new students or first-time visitors to the campus.

The existing workflow at COMSATS University Islamabad for campus guidance is entirely manual:

- **User Interaction:** Students, faculty, and visitors rely on physical notice boards, printed schedules, static maps, or staff assistance to locate classrooms, departments, or blocks.
- **Information Access:** Users must check multiple sources to find departmental details, faculty contacts, announcements, or event information.
- **Navigation:** Campus navigation depends on personal guidance or printed maps, with no interactive or real-time support.
- **Event & Announcement Updates:** Information about events and announcements is updated manually on notice boards, often leading to outdated or incomplete details.

- **Feedback Collection:** Feedback or suggestions are collected verbally or via paper forms and later reviewed manually.

The following figure, **Figure 2.1**, represents the **Current System Workflow** of campus guidance, providing an overview of the existing manual and offline processes, including physical inquiries, notice board consultations, printed map usage, and manual feedback collection.

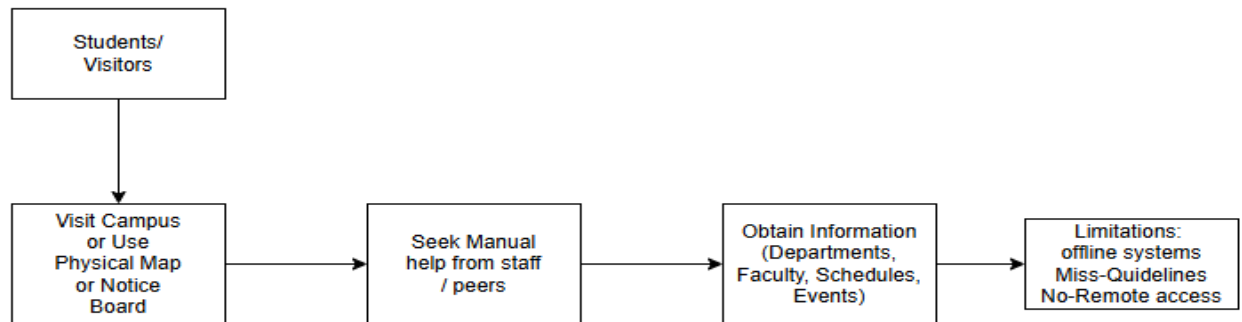


Figure 2.1 Current System Workflow of CUI

This manual model leads to delays in accessing information, frequent misunderstandings, and difficulty maintaining accurate records. Because all guidance depends on physical notice boards, printed maps, and verbal communication, users often face outdated details and inconsistent information. Transitioning to an online virtual guide will automate information delivery, provide real-time updates, ensure accuracy, and significantly improve accessibility for students, faculty, and visitors.

Chapter 3

Requirement Analysis

3. Requirement Analysis

Requirement Analysis is a critical phase in the system development process, aimed at understanding the needs and expectations of users and stakeholders, and converting them into a well-defined set of system requirements. For the CUI Virtual Guide, this phase focuses on identifying the essential features needed to assist students, faculty, and visitors in accessing campus information, navigating departments, viewing faculty details, and receiving real-time updates. It includes the examination of both functional and non-functional requirements, along with the development of use case diagrams to represent key user interactions within the virtual guidance system.

3.1. Use Case Diagram

A Use Case Diagram provides a high-level view of how users interact with the CUI Virtual Guide system. In this project, the main actor is the **User**, who may be a student, visitor, or faculty member seeking campus information, navigation support, event details, or assistance. The diagram captures all major system functionalities available to the user, such as viewing campus information, accessing navigation routes, scanning QR codes on kiosks, checking event schedules, registering for events, contacting support, and submitting feedback. Advanced features like the AI-powered campus assistant, AI-based voice assistant, location search, and smart route optimization further enhance the user experience by offering intelligent, real-time guidance across the campus. The relationships between use cases, including «include» and «extend» connections, show how detailed tasks such as viewing building information, getting route details, or updating the FAQ section are triggered within broader user actions. Figure 3.1 visually summarizes these interactions, illustrating how the CUI Virtual Guide system supports an efficient, automated, and user-friendly campus assistance process.

Actor

- **User** (Student, Faculty, Visitor)

Primary Use Cases

Campus Information

- View Building Information
- View Location Information

Navigation & Directions

- Get Navigation Directions
 - «include» View Campus Map
 - «include» Get Route Details

QR Code Scanning on Kiosk

- Display Information for Building

Event Information

- Get Event Information
 - «include» View Upcoming Events
- Register for an Event
 - «include» Submit Registration Details

Support & Communication

- Contact Support
 - «include» Submit Support Request
- Submit Feedback
 - «extend» Update FAQ Section

AI-Based Features

- AI-Powered Campus Assistant
 - «include» Get Navigation System
- AI-Based Voice Assistant
- AI-Based Location Search
 - «include» Search for a Specific Location
- Smart Route Optimization
 - «extend» Get Navigation Directions

USER USE CASES



Figure 3. 1 User Use Case Diagram for CUI Virtual Guide System

Actor

- **Admin (System Administrator, Campus Management Staff)**

Primary Use Cases

Campus Information Management

- **Manage Building Information**
 - «include» Add New Building
 - «include» Update Building Details
 - «include» Delete Building Entry

- **Manage Location Information**

- «include» Add Location
- «include» Edit Location Data
- «include» Remove Location

Navigation & Map Management

- **Update Campus Map**

- «include» Upload New Map Layout
- «include» Modify Existing Map Coordinates

- **Manage Route Information**

- «include» Add Route Details
- «include» Edit Route Data
- «extend» Verify Smart Route Optimization

QR Code & Kiosk Management

- **Generate QR Codes for Buildings**

- «include» Create QR for New Building
- «include» Update Existing QR Codes

- **Manage Kiosk Information Display**

- «include» Update Display Content
- «include» Sync Data With Firebase

Event Administration

- **Manage Events**

- «include» Add Upcoming Events
- «include» Edit Event Information
- «include» Delete Event

- **View & Approve Event Registrations**

- «extend» Notify Participants
- «include» Export Registration Data

Support & Feedback Handling

- **View Support Requests**

- «include» Respond to Support Queries
- «extend» Assign Issues to Departments

- **Manage Feedback**

- «include» Review Submitted Feedback
- «extend» Update FAQ Section

AI & System Management

- **Manage AI-Assistant Training Data**
 - «include» Update Navigation Dataset
 - «include» Add Location Keywords
- **Monitor System Analytics**
 - «include» View User Activity Logs
 - «include» Check System Performance
- **Manage User Roles**
 - «include» Add Staff Accounts
 - «include» Modify Access Roles
 - «include» Deactivate User Accounts

Figure 3.1 depicts the interactions between the admin and the CUI Virtual Guide system, showcasing core management and automation functionalities.

ADMIN USE CASES

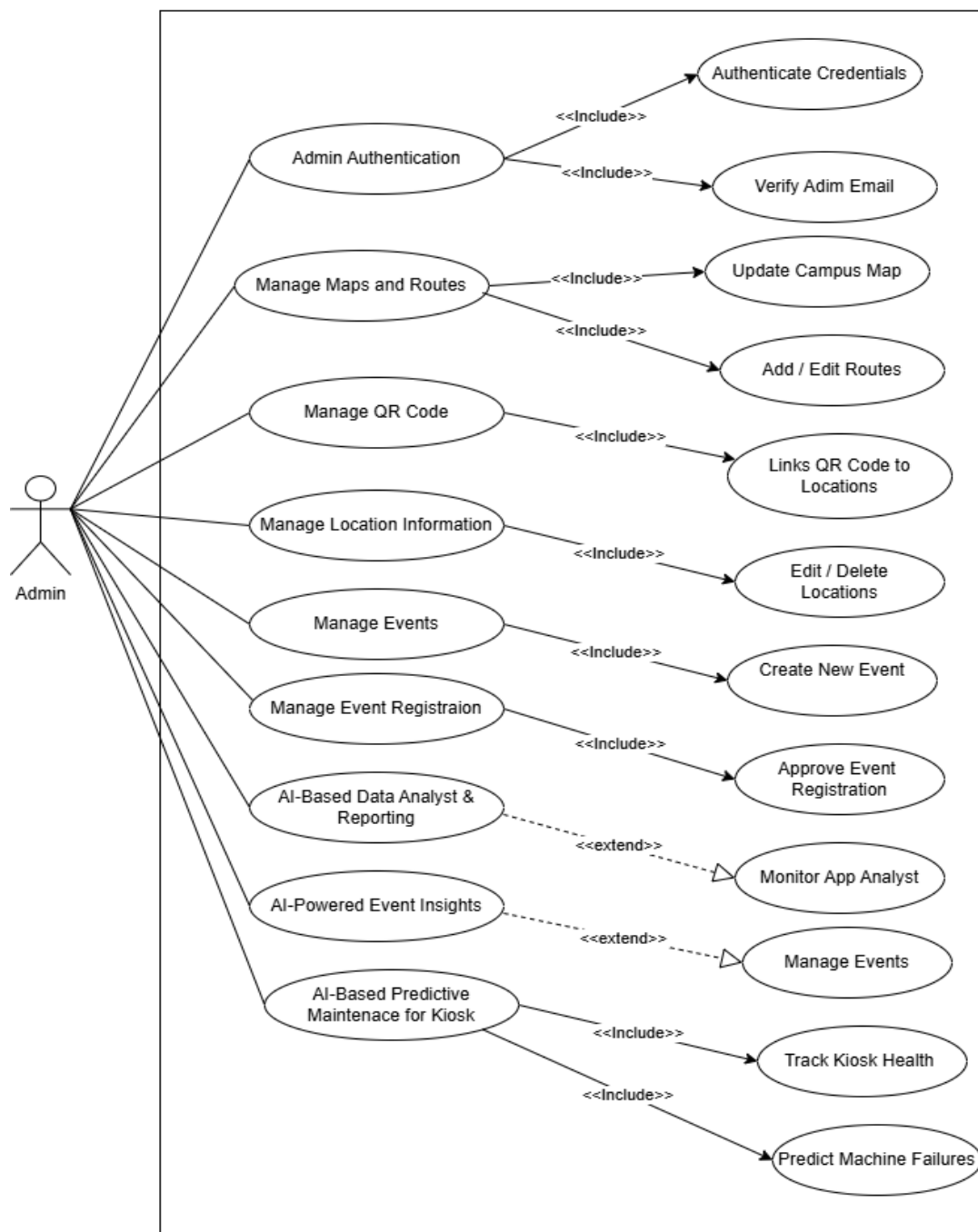


Figure 3. 2 Admin Use Case Diagram for CUI Virtual Guide System

3.2. Use Case Descriptions

Below is a table describing the key use cases identified for the **CUI Virtual Guide** system. Each use case outlines the involved actors, a brief description, and the overall flow of events. The following table illustrates the use case scenarios for the CUI Virtual Guide, categorized into two main sections: User (Student, Faculty, Visitor) and Admin (Campus Management Staff).

Use Case Name: User and Admin Roles**Actors: User (Student, Faculty, Visitor), Admin (Campus Management Staff)****Preconditions:**

- **For Users:**

- The user has access to the CUI Virtual Guide system through the mobile/web interface or campus kiosk.
- The user may optionally be logged in when accessing features like event registration or feedback submission.

- **For Admins:**

- The admin is authenticated in the system.
- Campus information (buildings, locations, events, routes, FAQs) exists and is stored in the Firebase database.

General Description:

This use case describes how **Users** interact with the CUI Virtual Guide to access campus information, navigate through buildings, view events, get AI-based assistance, and submit queries. It also describes how Admins manage and update the campus data, including building details, maps, events, routes, and support requests through the Firebase-backed admin console.

3.2.1. Main Flow:

The system displays the selected campus feature, whether it is a building description, event details, the campus map, or a navigation option. Based on this interface, the user chooses an action such as viewing more details, requesting directions, scanning a QR code, or registering for an event. Once the action is selected, the system retrieves all required information directly from the Firebase database to ensure accuracy and real-time updates. If the user requests navigation, the system automatically generates the best route and presents step-by-step directions on the screen. In the case of an event registration, the system records the user's submitted information in the database for future reference. Similarly, if a feedback message or support request is submitted, it is immediately forwarded to the admin panel for review. After completing any action, the system provides a clear confirmation message such as “Route

generated,” “Registration successful,” or “Feedback submitted,” ensuring the user knows their request has been processed.

3.2.2. User Use Cases

Table 3.1: User Use Cases outline the primary actions available to students, faculty, and campus visitors. These include viewing campus information, navigating buildings, scanning QR codes, checking events, registering for events, and seeking support.

These use cases ensure an interactive, smooth, and user-friendly campus guidance experience.

Table 3.1: User Use Case Details for "CUI Virtual Guide"

Use Case	Description
View Campus Information	Users can view detailed information about buildings, departments, labs, offices and facilities.
View Campus Map	Users can see the complete campus map with labeled locations.
Get Navigation Direction	Users can get step-by-step outdoor navigation routes to selected destinations.
Scan QR Code	Users can scan QR codes placed around campus to instantly view location information.
View Events	Users can browse upcoming campus events, seminars, and activities.
Register for Events	Users can register for campus events through the kiosk interface.
Submit Feedback	Users can share feedback or suggestions regarding their experience.
Contact Support	Users can submit help or support requests for any issue.

AI-Based Voice Assistance	Users can interact with the kiosk using AI-powered voice commands.
AI-Based Location Search	Users can search for any location using AI-enhanced search results.

3.2.3. Admin Use Cases:

Table 3.2 outlines the administrative use cases for managing the "CUI Virtual Guide" system. These include updating maps, managing events, handling QR codes, viewing user feedback, and generating AI-based analytic reports.

These use cases ensure smooth operation and continuous improvement of the kiosk system

Table 3.2 Admin Use Case Details for "CUI Virtual Guide"

Use Case	Description
Admin Authentication	Admin logs into the backend system to access management features.
Manage Maps & Routes	Admin updates campus map, modifies navigation paths, and corrects routes.
Manage QR Codes	Admin generates, updates, or links QR codes for campus locations.
Manage Location Information	Admin adds, edits, or removes building or departmental information.
Manage Events	Admin adds, updates, or deletes upcoming campus events.
Manage Event Registrations	Admin reviews and approves event registration requests.
Manage Support Requests	Admin checks and responds to users' support or help queries.
Manage Feedback	Admin reviews user feedback and updates FAQs if needed.

AI-Based Data Analysis & Reporting	Admin generates AI-powered insights about user interactions, routes, and usage patterns.
AI-Powered Event Insights	Admin views AI suggestions for event planning based on trends.
AI-Base Predictive Maintenance	Admin receives AI predictions for kiosk maintenance and health monitoring.

3.3. Functional Requirements

Functional requirements specify what the CUI Virtual Guide System must do to meet user needs and support campus navigation, information access, and administrative processes. For the CUI Virtual Guide, these requirements break down the core features and interactions that a user (student, faculty, visitor) or administrator will perform.

They include accessing campus information, viewing buildings and departments, retrieving navigation routes, scanning QR codes placed across campus, browsing events, and registering for them. On the administrative side, functionalities include managing campus maps, updating routes, generating and assigning QR codes, maintaining event records, analyzing usage data through AI, and managing locations or building information. Each functional requirement maps directly to the use case diagram, ensuring that every interaction (e.g., “Get Navigation Directions,” “Scan QR Code,” “Manage Events,” “AI Reporting”) is clearly defined, testable, and traceable throughout the development process.

- 1. FR1 – View Campus Information:** The system must allow users to view building details, department information, and location descriptions.
- 2. FR2 – Campus Navigation:** The system must provide turn-by-turn navigation, route details, and display the interactive campus map
- 3. FR3 – QR Code Scanning:** The system must allow users to scan QR codes and immediately view the mapped location information.
- 4. FR4 – Event Information Access:** The system must allow users to view upcoming events with dates, descriptions, and venue details.

- 5. FR5 – Event Registration:** The system must allow users to register for events by submitting required details through the kiosk.
- 6. FR6 – Feedback & Support Submission:** The system must allow users to submit complaints, queries, or suggestions using a feedback form.
- 7. FR7 – AI-Based Assistance:** The system must use AI to provide voice-based navigation help, location search, and intelligent suggestions.
- 8. FR8 – Admin Authentication:** The system must authenticate administrators securely for accessing the management dashboard.
- 9. FR9 – Manage Campus Maps & Routes:** The system must allow administrators to update map data, modify routes, and add new locations.
- 10. FR10 – Manage QR Codes:** The system must allow administrators to generate, assign, or update QR codes linked to campus locations.
- 11. FR11 – Manage Events:** The system must allow administrators to add, edit, or delete event records.
- 12. FR12 – AI-Based Analytics & Reporting:** The system must provide administrators with AI-generated reports, including system usage insights, event trends, and kiosk activity.

3.4. Non-Functional Requirements

Non-functional requirements describe how the CUI Virtual Guide System performs rather than what it does, focusing on quality attributes that impact system efficiency, user satisfaction, reliability, and long-term maintainability. For the CUI Virtual Guide, performance requirements ensure fast loading of maps, events, and navigation routes to support a smooth user experience on the kiosk. Usability requirements emphasize a simple, intuitive interface that enables students, visitors, and faculty to access essential campus information in minimal steps.

Security measures ensure that administrative data such as event records, QR code mappings, and map configurations is protected through secure authentication and controlled access. Scalability requirements ensure that future features such as additional AI capabilities, integration with campus databases, or IoT-based sensors can be added without restructuring

the system. Maintainability and compatibility guidelines ensure clean code, clear documentation, and optimal performance on kiosk devices, Android tablets, or touchscreen displays used on campus.

1. NFR1 – Performance: The system must load maps, building information, and navigation routes within two seconds on kiosk hardware under normal usage.

2. NFR2 – Usability: The interface must be simple and intuitive, enabling users to access key features (such as viewing locations or getting directions) in no more than three taps/clicks.

3. NFR3 – Security: Admin access must be secured using authentication, and sensitive data (routes, QR mappings, event details) must be stored and transmitted securely.

4. NFR4 – Scalability: The system architecture must support future integration of AI models, new map layers, additional kiosks, and campus-wide extensions without requiring major redesign.

5. NFR5 – Compatibility: The system must run smoothly on Android-based kiosks, touchscreens, tablets, and modern browsers, ensuring consistent performance across supported devices.

6. NFR6 – Maintainability: The codebase must follow best practices (modular design, clean structure, and documentation), allowing administrators or developers to update maps, routes, or AI modules easily.

Chapter 4

Design and Architecture

4. Design and Architecture

This chapter presents the core system architecture, data organization, workflow structure, and design models of the **CUI Virtual Guide System**. The architectural design defines how different components such as the kiosk interface, navigation module, QR code system, event management module, and admin dashboard interacts to deliver a seamless and efficient user experience.

The system's design focuses on scalability, maintainability, and reliability, ensuring smooth operation on campus kiosks and compatibility with future expansions, including AI-powered features. Each architectural component aligns directly with the functional and non-functional requirements defined in earlier chapters, enabling a structured approach to system development and deployment.

4.1. System Architecture

The overall architecture of the **CUI Virtual Guide** follows a **multi-layered system model**, separating the platform into the User Interface Layer, Application Layer, Data Layer, and Integration Layer. This layered design ensures modularity, supports future enhancements such as AI-based navigation assistance, and maintains smooth interaction between the kiosk system, mobile app, and university backend services.

4.1.1. User Layer (Kiosk + Mobile App Frontend)

The user interacts with the system through:

- Kiosk Interface (Touchscreen) placed at the university entrance or key locations.
- Mobile Application that uses QR Code scanning to access navigation and event details.

Provides an intuitive and responsive UI for:

- Campus navigation
- Event viewing
- Voice-based assistance
- QR-based information retrieval

Ensures accessibility and ease of use for students, visitors, and faculty.

4.1.2. Application Layer (Core Functional Logic)

- Contains all essential processing components of the system:
- User Interface Controllers manage screen transitions and input handling for both kiosk and mobile app.
- Navigation Module calculates routes and displays directions based on campus map data.
- Business Logic Services manage rules for event scheduling, location details, and content rendering.
- Voice Engine converts text instructions into speech for kiosk-based voice interaction.
- Event Handler Module manages the event calendar, reminders, and real-time updates.
- QR Code Generator & Scanner Module produces dynamic QR codes for buildings/events and decodes scanned codes from the app.
- Ensures that all UI requests are processed efficiently and accurately.

4.1.3. Data Layer (Databases & Storage)

- Maintains structured datasets required by the system:
- Map Database stores all campus locations, pathways, departments, and building coordinates.
- Event Database includes academic schedules, seminars, and university event information.
- Logs & Stats track user interactions, frequent search queries, and system performance metrics.
- Ensures data consistency and quick access for navigation and event retrieval.
- All databases are organized to support scalability for future campus expansions or new features.

4.1.4. Integration Layer (External APIs)

The Integration Layer connects the CUI Virtual Guide to external services that enhance system functionality without increasing internal complexity.

- Provides access to official campus data such as department details, announcements, events, and general campus information. This ensures the Virtual Guide displays accurate and updated content.

- Enables the chatbot to understand user voice commands. Spoken queries from users are converted into text, which the system then processes to generate appropriate responses. This improves accessibility and creates a more interactive experience.
- The mobile application uses an integrated QR scanning library (such as ML Kit or ZXing) to scan campus QR codes. This allows users to quickly access building information, navigation guidance, and event details.

4.1.5. Communication Flow:

- The kiosk or mobile app initiates user actions such as navigation, QR scanning, or viewing events.
- Requests are sent to the Application Layer, where controllers process them and trigger appropriate modules.
- The Application Layer accesses the Data Layer for retrieving map or event data.
- Responses are returned back to the kiosk or mobile app interface in the form of:
 - Visual navigation routes
 - Voice instructions
 - Event details
 - QR-based content

Figure 4.1 provides a high-level overview of the CUI Virtual Guide system architecture and its major functional components.

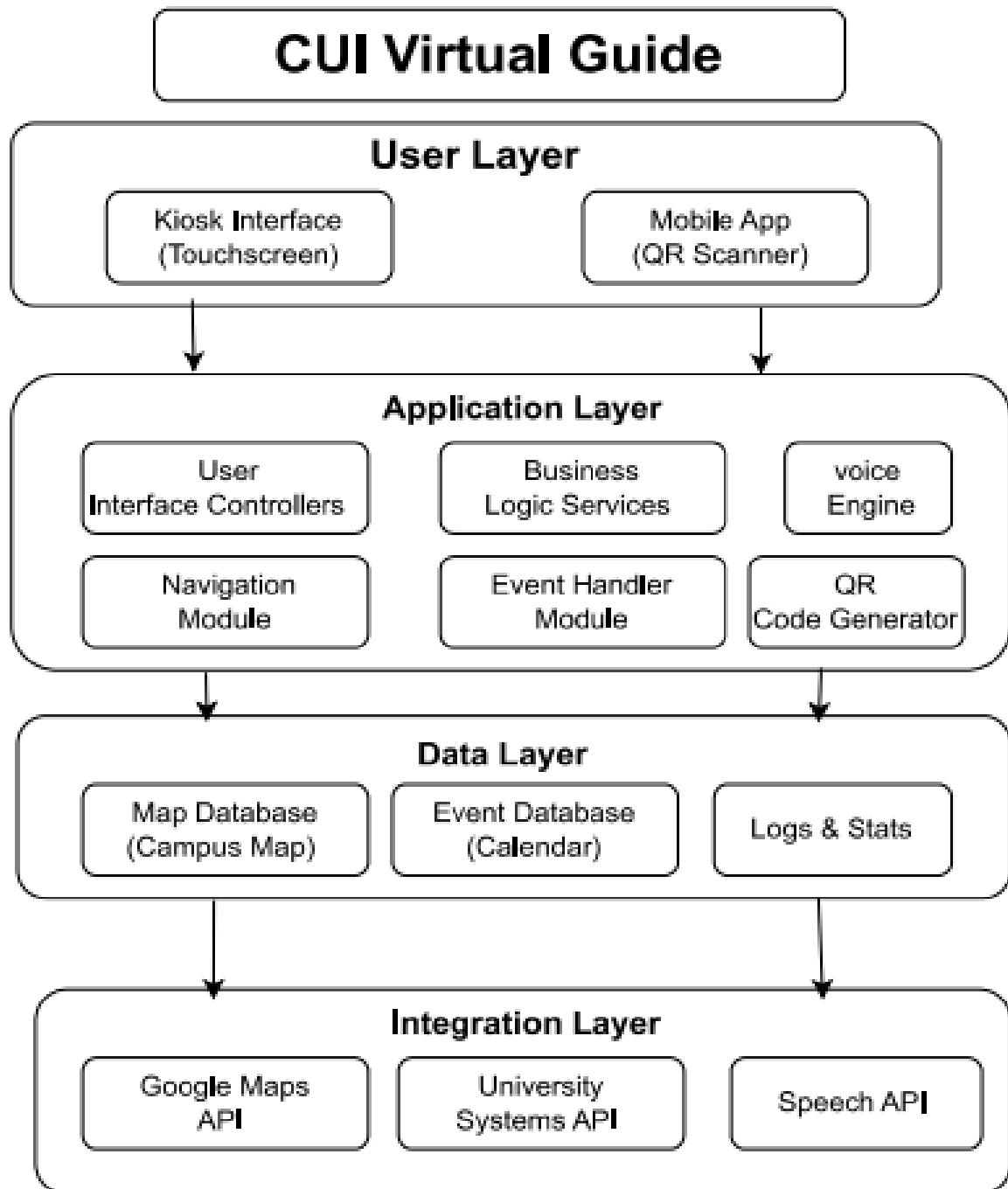


Figure 4.1 System Architecture Diagram of CUI Virtual Guide

4.2. Process Flow/Representation.

The system's process flow begins with user interaction, where individuals access the kiosk and select their preferred language or role. Once initialized, users can search for locations using the Navigation Engine, scan or generate QR codes for directions or information using the Manager, and use the Voice Assistant to issue spoken commands or receive audible

responses. As events are scheduled, the Event Manager provides users with details and allows them to filter by date. Users can also provide feedback through the Feedback System, which processes sentiments and creates support tickets if needed. In the background, administrators utilize the Analytics Dashboard to monitor system performance, track usage metrics, and generate reports

4.2.1. Main System Flow

Figure 4.2 shows the following:

- System Startup: Kiosk powers on and loads core modules.
- Home Screen Initialization: User interface loads and displays the main menu.
- User Interaction: User selects an option using touch or speech input.

Module Selection

Based on the user's choice, the system redirects to one of the following:

Navigation Module:

Processes destination queries and displays route guidance or QR-based navigation.

Event Information Module:

Retrieves and shows upcoming campus events and announcements.

QR Module:

Scans or generates QR codes and displays related information.

Voice Assistant:

Converts speech to text and executes recognized commands.

Feedback System:

Collects user comments and stores them as feedback entries.

Return to Home

After completing a task, the system returns to the home screen for further interaction.

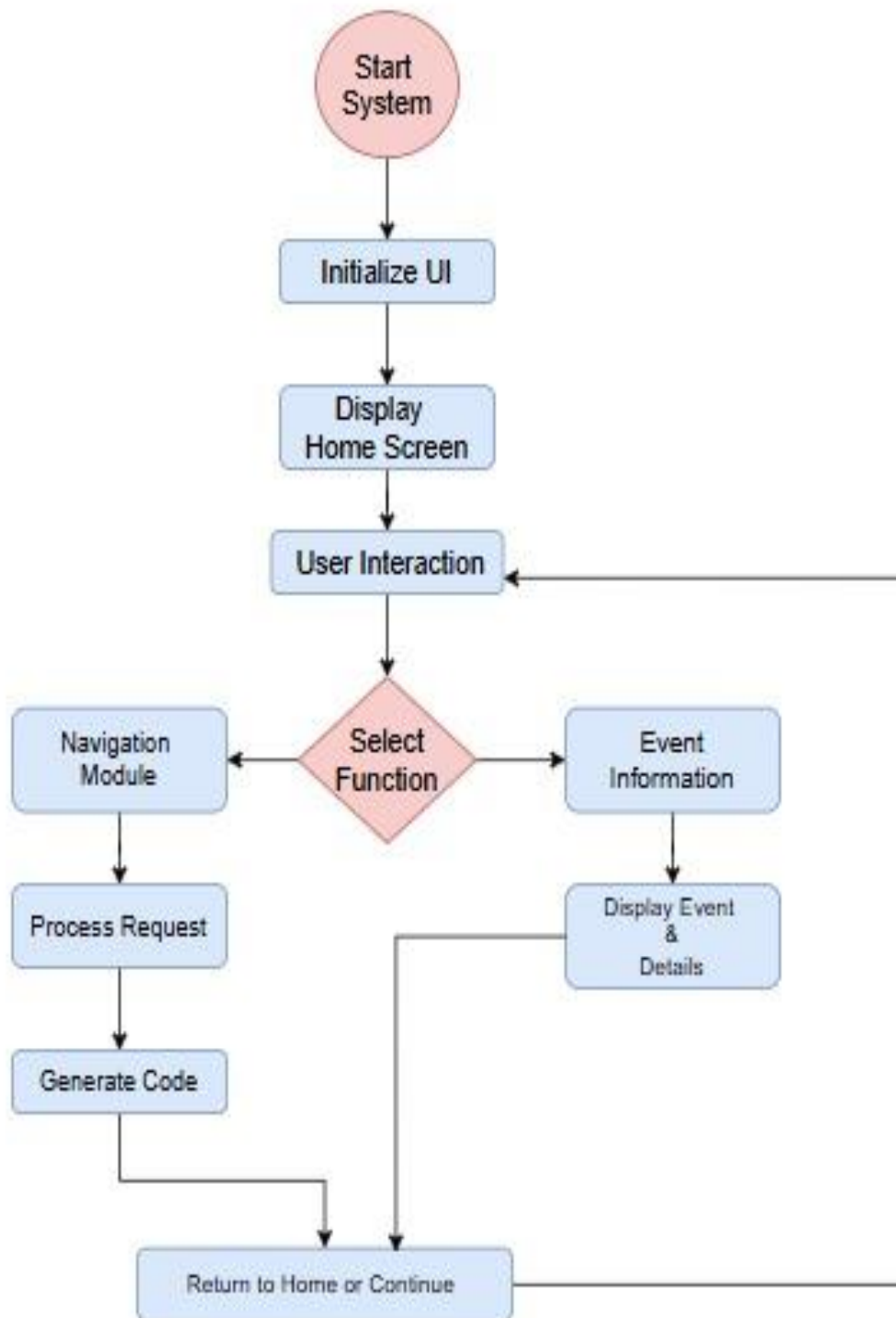


Figure 4.2 Main System Flow

4.2.2. Navigation Module Flow

Figure 4.3 shows a flowchart that illustrates the steps involved in a navigation system process. It begins with starting navigation and selecting a destination, followed by querying a database for a map. The chart includes a decision point to check if the location is found, either

proceeding to calculate and display the route or displaying an error, before finally returning to the main menu.

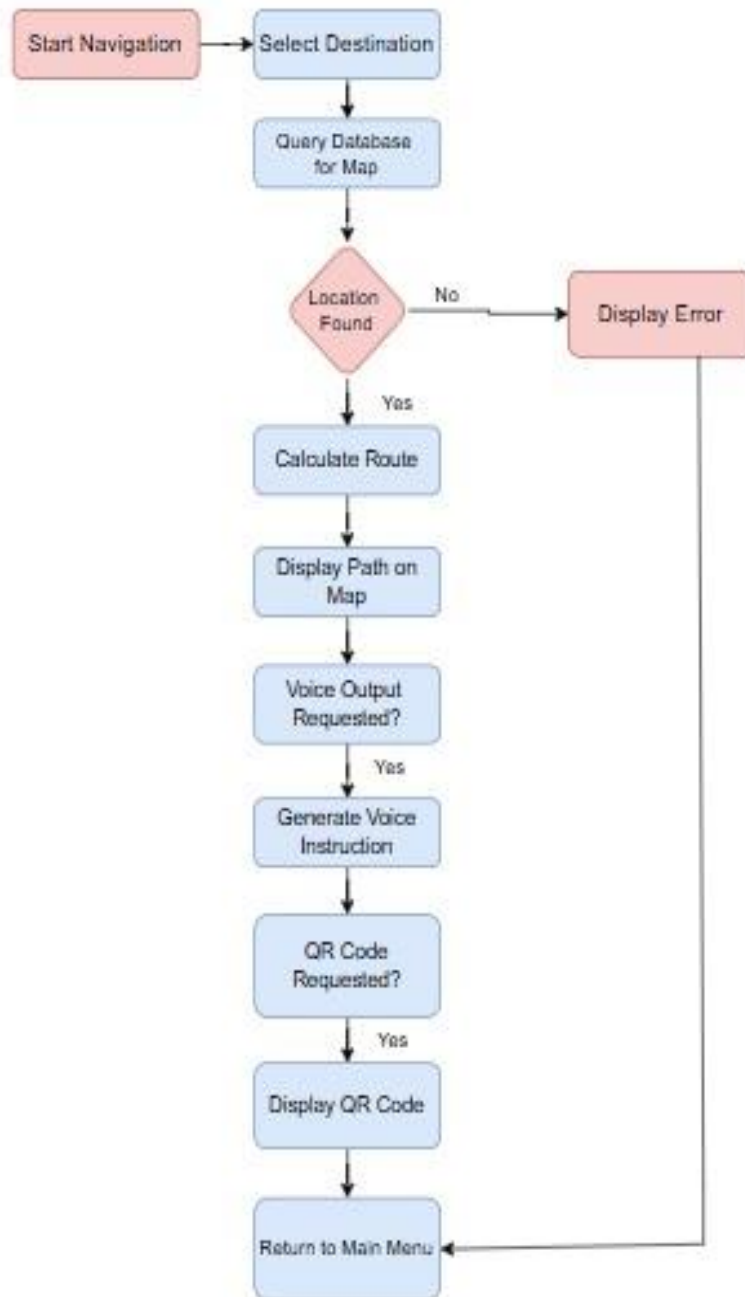


Figure 4.3 Navigation Module Flow

4.2.3. Feedback Module Flow

Figure 4.4 shows a flowchart outlines the steps for a Feedback Process, starting with displaying a form where the user inputs a rating and comments. It includes a validation

check to either save the feedback to a database and display a thank you message, or show an error message, before concluding by returning to the main menu.

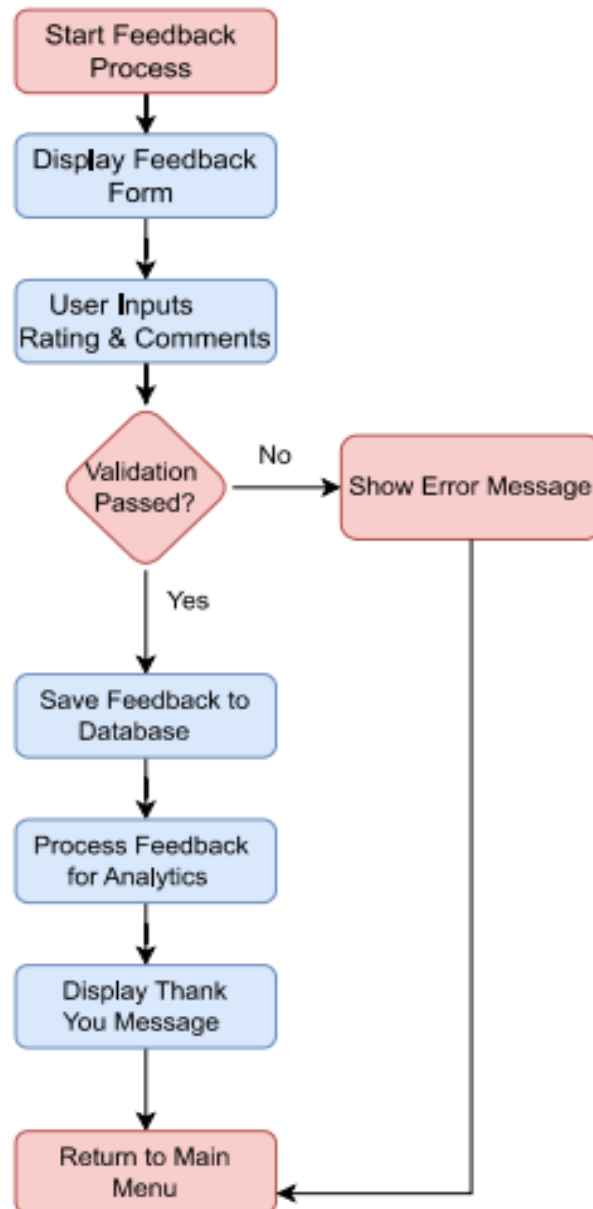


Figure 4.4 Feedback Module Flow

4.2.4. Voice Assistant Flow

Figure 4.5 shows a flowchart illustrates the core process of a Voice Assistant, beginning with activation and listening for a command. It uses Natural Language Processing (NLP) to identify the user's intent, and then either generates an appropriate voice response by routing to a specific module or handles errors/requests clarification if the speech or intent cannot be recognized.

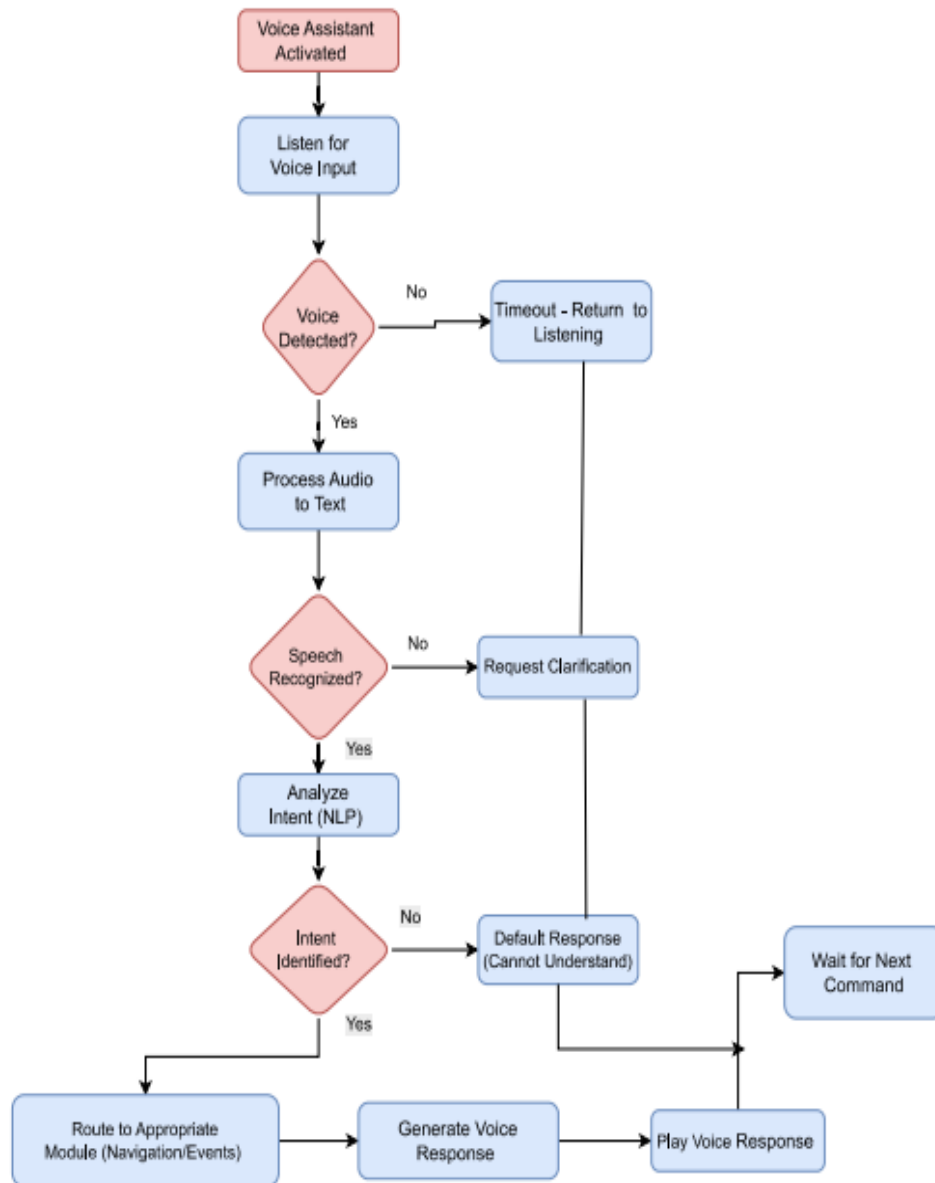


Figure 4.5 Voice Assistant Flow

4.3. System Modules

The CUI Virtual Guide system is structured into specialized modules, each responsible for a specific functionality within the kiosk platform. This modular architecture ensures independence, easier maintenance, and smooth integration across all features.

1. User Interface Module

Handles all visual and interactive components of the system, such as:

- Welcome screen
- Main menu
- Navigation buttons
- Event information pages

- Feedback
- Ensures a clean and touch-friendly UI experience.

4. Navigation & Location Module

Provides directional assistance and campus guidance.

Includes:

- QR-based navigation
- Department locations
- Static and interactive map views
- Routes to frequently visited areas

3. Events & Information Module

- Displays up-to-date university information, including:
- Campus announcements
- Upcoming events
- Department

All content is fetched from the backend and displayed in a structured format.

4. QR Module

Handles:

- QR code scanning to fetch relevant data
- QR code generation for navigation and quick access

5. Chatbot Module

Provides voice-based assistance through:

- Speech input
- Speech-to-text conversion
- Local processing of user queries
- No external AI API is used; processing occurs within the system.

6. Feedback Module

Allows users to submit

- Suggestions
- Queries
- Complaints

Feedback is stored in the backend for administrative review.

7. Database Management Module

Responsible for storing and managing backend data including:

- Events
- Locations
- Departments
- Feedback entries

8. Admin Module (Replaces System Module)

Handles administrative operations such as:

- Adding/updating events
- Managing location data
- Uploading department details
- Reviewing and responding to user feedback
- Updating QR-linked information

The Admin Module ensures the system remains updated and consistent across all feature.

4.3. Design Models

The design models describe both the structural and behavioural aspects of the CUI Virtual Guide system. These models act as blueprints for developers and provide a clear understanding of how different modules and components interact within the application. They illustrate how the system behaves over time and how various elements are organized internally. For this project, the design models present a detailed view of user interactions, system workflows, and module communication. Multiple sequence diagrams are included to explain the step-by-step flow of user actions such as navigation requests, event information retrieval, QR scanning, and chatbot queries. Additionally, an overall class diagram is provided to represent the core architecture of the system, showing key classes, their attributes, methods, and interrelationships. These models help developers, testers, and stakeholders visualize how the Virtual Guide operates, ensuring clarity, consistency, and efficient.

4.4.1. Sequence Diagram

The sequence diagram outlines user interaction with the kiosk system and its components. It begins with language or role selection, followed by actions like location search.

4.4.1.1. Sequence Diagram of Campus Navigation

Figure 4.6 shows how the CUI Virtual Guide generates a navigation route. The user selects a destination on the Kiosk UI, which sends a route request to the Navigation Module. The module queries the database for map data, calculates the optimal route, and returns it to the Kiosk UI. The kiosk then displays the route and generates a QR code, which the user can scan to view the path on their mobile device.

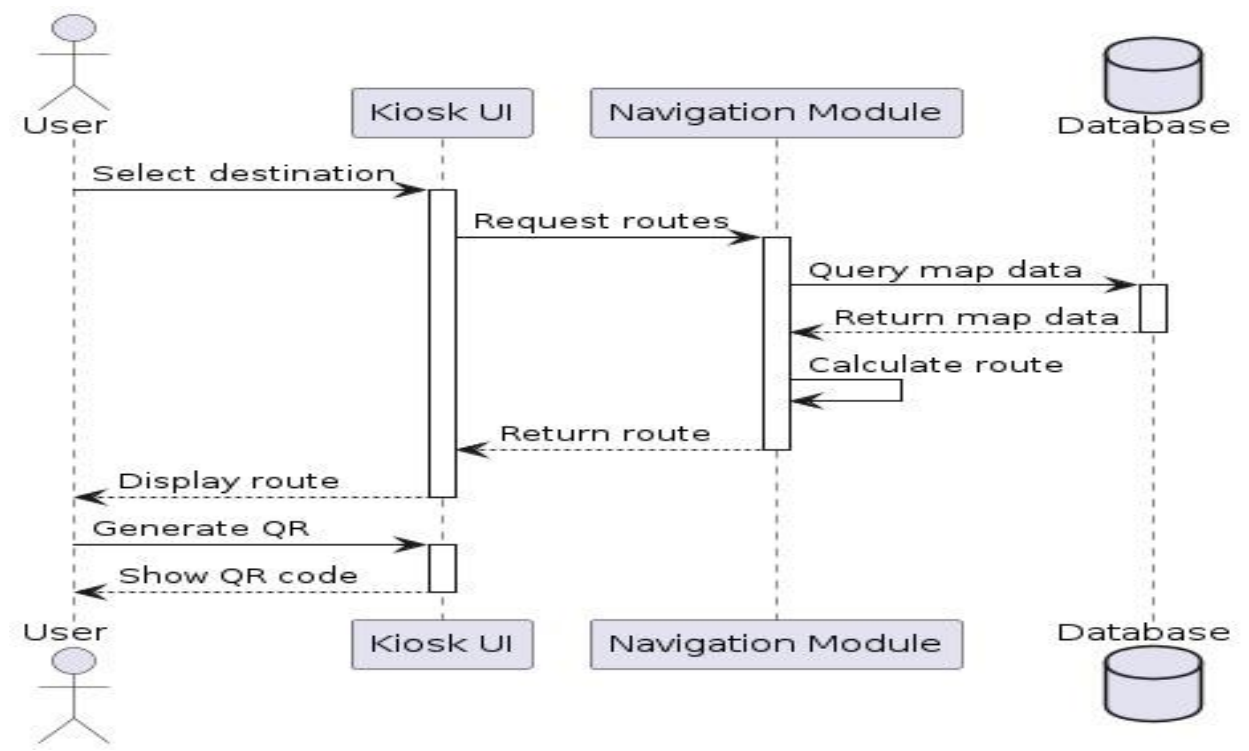


Figure 4.6 Sequence Diagram – Campus Navigation

4.4.1.2. Sequence Diagram of Voice Assistant

Figure 4.7 shows a sequence diagram showing how a user's voice query is processed by a system. The flow demonstrates the interaction between the User, Voice Module, NLP Processor, and Response Generator to process the audio, extract intent, generate a response, and finally speak it back to the user.

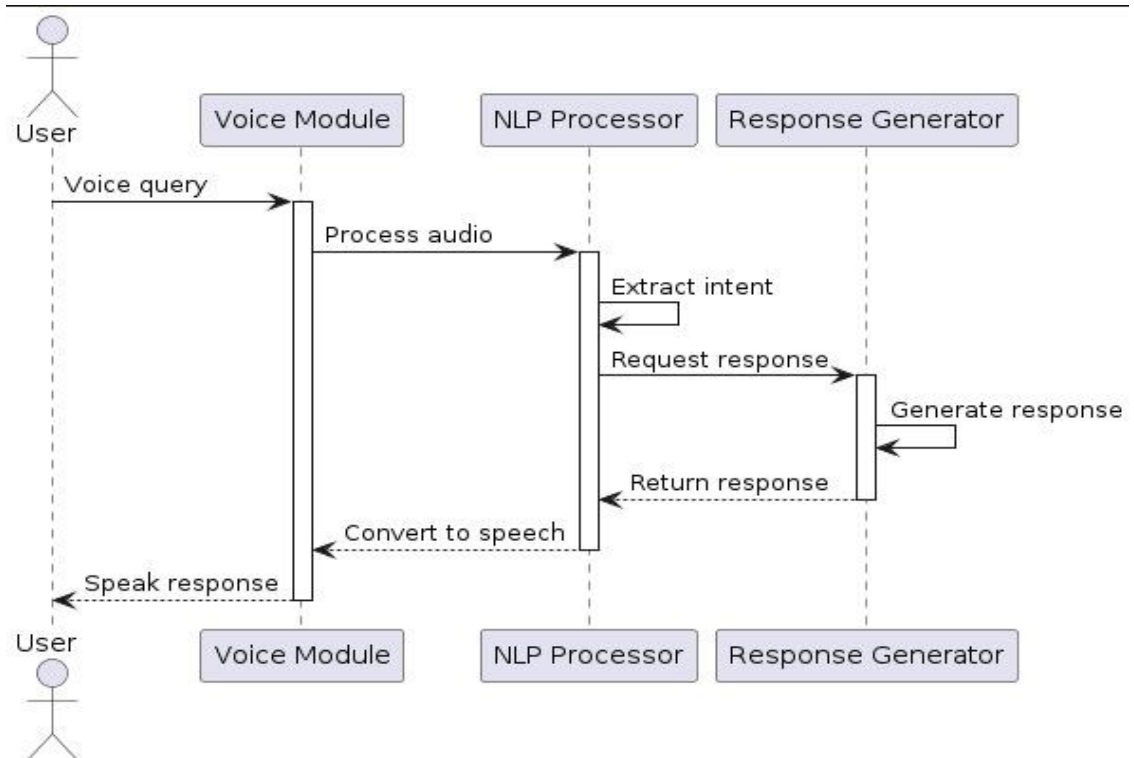


Figure 4.7 Sequence Diagram of Voice Assistant

4.4.1.3. Sequence Diagram of QR Code Integration

Figure 4.8 shows a sequence diagram illustrates the process of transferring route information from a Kiosk to a user's mobile device via a QR code. The Kiosk UI requests and displays a QR code generated from the route details, which the User then scans with their Mobile Device to process and display the route.

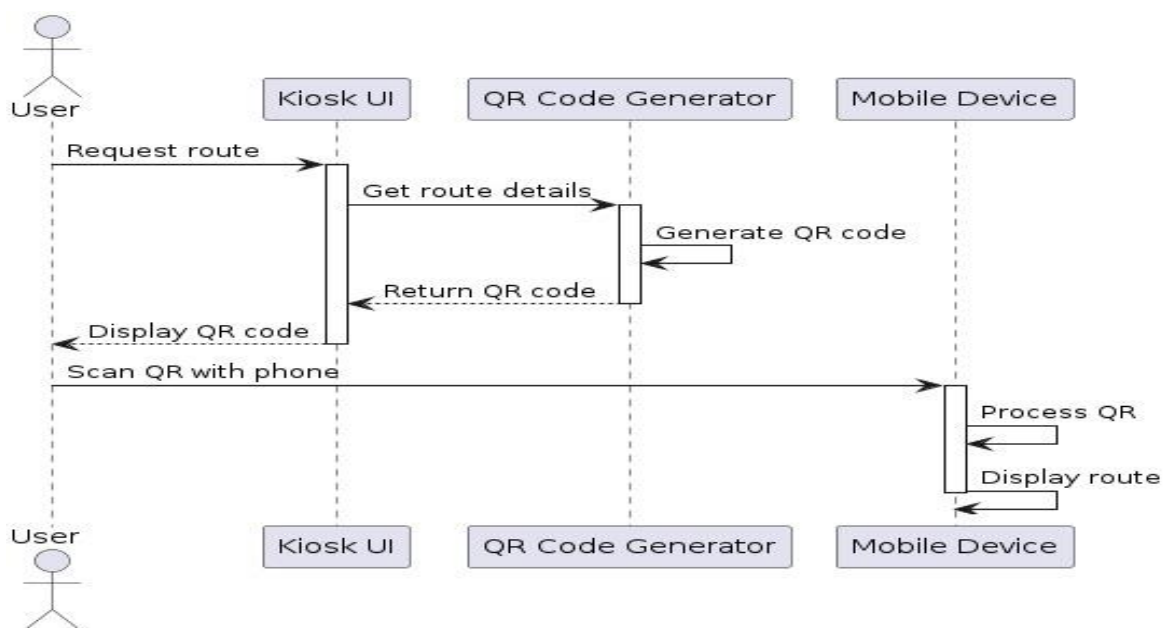


Figure 4.8 Sequence Diagram of QR Code Integration

4.4.1.4. Sequence Diagram of Event Information

Figure 4.9 shows the steps where a User requests events via the Kiosk UI, which triggers the Event Module to query the Event Database, process the data (filter/sort), and then display the returned event list and details to the user.

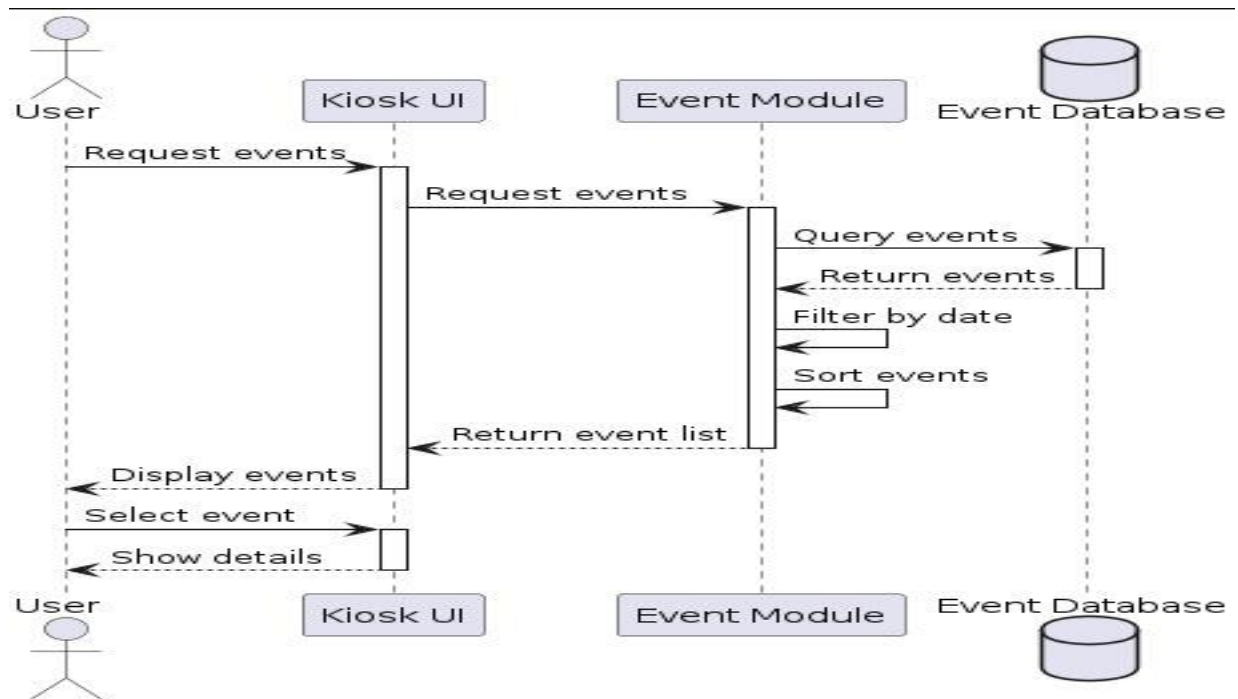


Figure 4.9 Sequence Diagram of Event Information

4.4.2. Class Diagram

Figure 4.10 diagram provides a static view of the core components of the CUI Virtual Guide system and how they interact. It outlines the main classes responsible for navigation, map handling, user interaction, and administrative management. These structural relationships guide developers in implementing object-oriented logic across the system.

Relationships:

- NavigationModule → MapDatabase: Uses database data to calculate routes (dependency).
- KioskUI → NavigationModule: Sends user requests and receives route responses (association).
- NavigationModule → QRGenerator: Requests QR creation for the generated path (dependency)
- Admin → MapDatabase: Admin can update or modify campus/map-related data (association).
- KioskUI → SpeechModule: Converts user voice input into text commands (dependency).

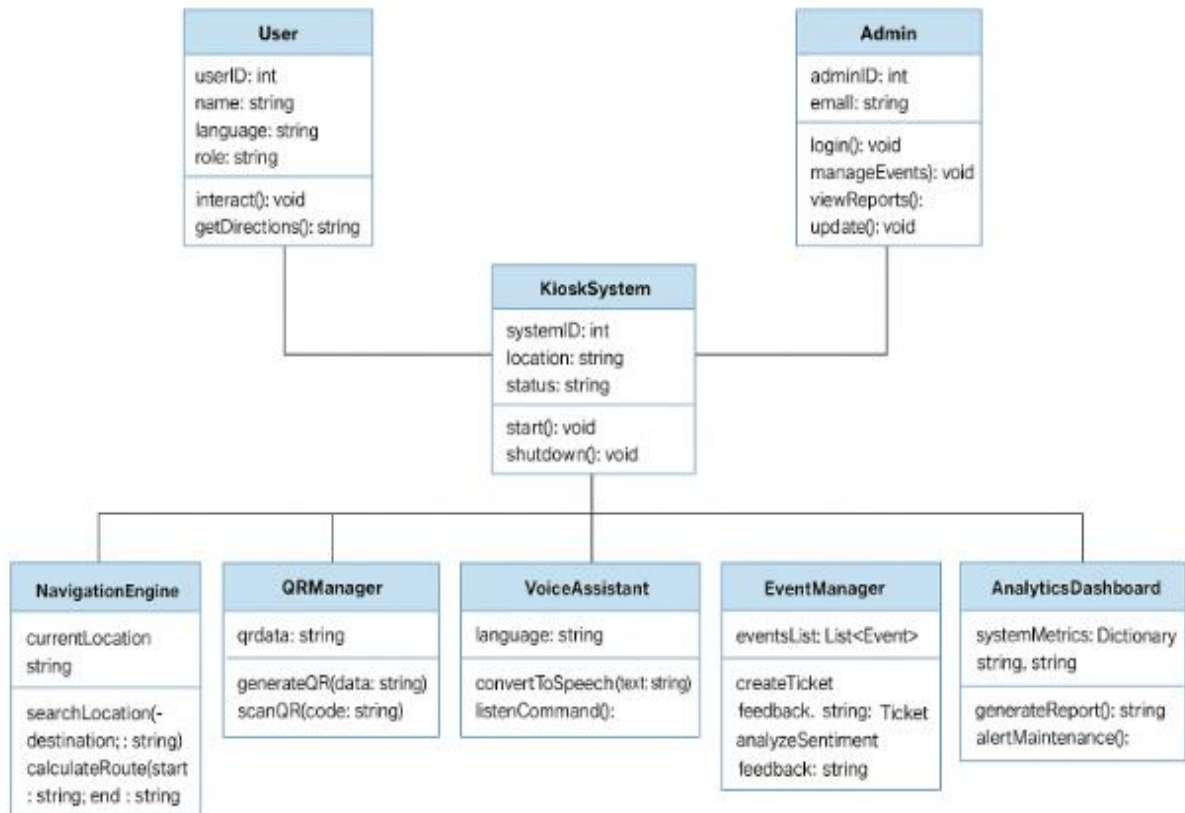


Figure 4.10 Class Diagram – System Structure

Chapter 5

Implementation

5. Implementation

This chapter outlines the implementation details of the CUI Virtual Guide project, describing the algorithms used, system modules developed, database structure, and the core functionalities integrated into the kiosk system. It also references UML diagrams that visually represent the system's architecture, workflow, and interaction between components.

The implementation of the CUI Virtual Guide is the result of systematic planning, detailed design, and the integration of intelligent features such as an AI-powered chatbot and an interactive campus navigation system. This chapter explains how the system was practically developed including the logic behind the navigation module, the data flow for event and announcement updates, the admin management features, and the interaction design used to create an intuitive user experience.

By covering algorithms, data processing, and UI development, this chapter bridges the gap between initial system architecture and the final functional kiosk system. Each section provides a clear explanation of the purpose, scope, and contribution of the individual components to the working of the overall CUI Virtual Guide.

5.1. Algorithm

The system's behaviour and logic are driven by core algorithms designed to ensure accuracy, smooth navigation, and efficient access to campus information. These algorithms define how the kiosk processes user input, retrieves relevant data, and displays results across key modules such as campus navigation, event information, departmental details, and chatbot responses. The core functionalities of the CUI Virtual Guide are supported by several key algorithms that manage system operations. Below is the pseudocode-style or natural language explanation of how major system features work:

5.1.1. Admin Login Algorithm

Function: adminLogin(email, password)

Input:

email (string)

password (string)

Output:

loginStatus (boolean – success or failure)

Pseudocode

1. Admin enters email and password in the login form.
2. Fetch the stored hashed password from Firebase for the given email.
3. Hash the entered password using the same hashing method (e.g., SHA-256/bcrypt).
4. Compare the newly hashed password with the stored hashed password.
5. If both hashes match → authentication successful → allow admin access.
6. If the hashes do not match → deny login and show an error message.
7. Log successful login event into the system for security tracking.

5.1.2. Navigation Algorithm

Predefined Route Definition Algorithm

Function: loadRoute(destination)

Input:

destination (string) – selected campus location

Output:

routePolyline (list of coordinate points)

Pseudocode

1. Receive selected destination from the user interface.
2. Match the destination with its predefined route stored in the project files.
3. Retrieve the corresponding list of polyline points (lat–lng pairs).
4. Clear any previously drawn route on the map.
5. Draw the retrieved polyline on the campus map using the map renderer.
6. Highlight the active route for visibility.
7. Return the generated routePolyline for display.

5.1.3. QR Code Scan Algorithm

QR-Based Screenshot Sharing Algorithm (using ImgBB Upload + ZXing QR Generation)

Function: generateQRCode(mapScreenshot)

Input:

mapScreenshot (bitmap image)

Output:

qrImage (bitmap)

Pseudocode

1. Capture the current map view as a bitmap.
2. Convert bitmap to JPEG format.
3. Encode JPEG image into Base64 string.
4. Send Base64 string to ImgBB API using POST request.
5. Receive hosted image URL from API response.
6. Encode URL into QR format using ZXing library.
7. Generate a QR bitmap from the encoded data.
8. Display QR code to user.
9. Return QR bitmap.

5.1.4. Feedback Submission

The system uses Firebase to store user-submitted contact messages.

Function: submitFeedback(name, email, message)

Input:

- name (string)
- email (string)
- message (string)

Output:

status (string) – "Success" or "Error"

Pseudocode

1. Receive input values (name, email, message) from the UI.
2. Validate fields (check empty, check valid email format).
3. If validation fails → return error message.
4. If validation passes → create a feedback object.
5. Store feedback object in **Firebase "Feedback" node/collection**.
6. Return success confirmation message to the user.

5.1.5. Contact Form Submission

Function: submitContact(name, phone, email, message)

Input:

- **name** (string)
- **phone** (string)
- **email** (string)

- **message** (string)

Output:

- **status** (string)

Pseudocode

1. Receive user inputs (name, phone, email, message) from UI.
2. Validate inputs (non-empty fields, valid phone number, valid email).
3. If validation fails → return error message.
4. Create a contactMessage object.
5. Store the object in Firebase "Contact_Messages" node/collection.
6. Return success confirmation to the user.

5.1.5. Contact Form Submission

Function: submitContact(name, phone, email, message)

Input:

- **name** (string)
- **phone** (string)
- **email** (string)
- **message** (string)

Output:

- **status** (string)

Pseudocode

1. Receive user inputs (name, phone, email, message) from UI.
2. Validate inputs (non-empty fields, valid phone number, valid email).
3. If validation fails → return error message.
4. Create a contactMessage object.
5. Store the object in **Firestore "Contact_Messages" node/collection.**
6. Return success confirmation to the user.

5.1.6. AI Voice Assistant Interaction

Function: processVoiceCommand(command)

Input:

command (string) – spoken user query converted into text

Output:

response (string) or **action** (navigation, information display)

Pseudocode

1. Start voice recognition and capture user speech input.
2. Convert speech to text using built-in Android SpeechRecognizer.
3. Send recognized text (command) to the processing module.
4. Match command with predefined intents (e.g., “find block”, “show map”).
5. If command matches navigation intent → extract destination keyword.
6. If command matches informational intent → fetch relevant data from Firebase.
7. Generate appropriate response (text + optional audio output).
8. Perform required action (e.g., open screen, show route, provide info).
9. Return processed response to the user.

5.2. External APIs

The CUI Virtual Guide Kiosk integrates three main external APIs/libraries to support essential operations such as QR generation, image hosting, voice interaction, and cloud-based data storage. Since the system uses a static campus map with predefined polylines, no Google Maps API is used. The external APIs ensure smooth navigation, interactive features, and efficient backend communication.

5.2.1. Text-to-Speech (TTS) API

The Text-to-Speech (TTS) API allows the AI Voice Assistant to convert written text into natural-sounding voice output. This feature helps users who prefer listening instead of reading and supports accessibility for individuals with reading difficulties. It improves user engagement by providing real-time spoken instructions, greetings, building details, and directions. By delivering information through audio, the system becomes more interactive and user-friendly for all types of users.

5.2.2. ZXing QR Code Generation API

The ZXing QR Code Generation API is used in the system to convert URLs into scannable QR codes. This allows users to receive map screenshots on their mobile phones by simply scanning the generated QR code displayed on the kiosk. The library is fast, lightweight, open-source, and fully compatible with common mobile QR scanners, including Google Lens. By enabling users to take the map with them, the system enhances mobility and ensures that navigation can continue even after leaving the kiosk.

5.2.3. Firebase Realtime Database / Firestore API

Firebase Realtime Database or Firestore is used to store feedback and contact form submissions from kiosk users. It allows users to submit feedback without needing to log in, while administrators can easily view all submissions directly through the Firebase console. As a secure and scalable cloud-based solution, Firebase removes the need to build a custom backend or maintain a server. It automatically synchronizes data in real time, making feedback handling simple, efficient, and reliable.

5.2.4. ImgBB Image Upload API

Table 5.1 shows a ImgBB Image Upload API is used to upload map screenshots and generate publicly accessible image URLs. It offers free cloud image hosting and uses a simple Base64 image upload process through a POST request. Once uploaded, ImgBB returns a direct URL that can be embedded into QR codes for user download. This eliminates the need for a database to store images and ensures quick, easy sharing of campus map screenshot

Table 5.1 Details of APIs used in the project

Name of API	Description of API	Purpose of Usage	List of Functions/Classes Used
Text-to-Speech (TTS) API	Converts text into spoken audio output.	Provides voice guidance and AI assistant interaction.	TextToSpeech, speak()
ZXing QR Generation API	Generates QR codes from text or URLs.	Creates QR codes for sharing map screenshots.	MultiFormatWriter, BitMatrix
Firebase Realtime Database / Firestore	Cloud-based database for storing structured data.	Stores feedback and contact submissions from kiosk users.	DatabaseReference, setValue()
ImgBB Upload API	Uploads images and provides public URLs.	Uploads map screenshots to generate shareable links.	OkHttpClient, Base64 encoding, POST request

5.3. User Interface

The user interface (UI) of the CUI Virtual Guide is designed to be simple, interactive, and easy for visitors, students, and staff to use. Since the kiosk serves as a digital guide for campus navigation, the interface focuses on clarity, accessibility, and smooth interaction. The goal is

to ensure that users can quickly find directions, explore buildings, scan QR codes, and interact with the AI Voice Assistant without confusion.

The UI includes a responsive and visually clean layout that displays essential navigation buttons, campus map access, department information, and a voice assistant trigger button. Interactive elements such as touch-based navigation panels, search options, and clear icons make the system easy to operate even for first-time users. For administrative use, the system provides feedback viewing interface where admins can access submitted feedback and contact form entries stored in Firebase. Overall, the UI ensures that both visitors and admin users can interact with the kiosk smoothly and efficiently.

5.3.1. Welcome Screen

Figure 5.1 Welcome Screen of the CUI Virtual Guide app presents a visually appealing and professional interface, set against a vibrant background image of the COMSATS University Islamabad, Sahiwal Campus. At the top of the screen, the university's official logo and name are clearly displayed, establishing authenticity and a strong institutional identity. Centered in bold white text is a warm welcome message that reads **“WELCOME CUI Virtual Guide”** instantly communicating the app's purpose. Below the message, a prominent blue **“START”** button invites users to begin their journey, encouraging interaction with the app's main features. The overall design is clean, inviting, and informative, offering a smooth and welcoming introduction to first-time users as they prepare to explore the campus virtually.



Figure 5.1 Welcome Screen

5.3.2. Main Screen

Figure 5.2 Main Screen of the CUI Virtual Guide app offers users an interactive and feature-rich interface designed to assist in campus navigation and information access. Dominating the centre of the screen is a detailed campus map, enhanced with a compass and location marker to help users orient themselves easily. A vertical sidebar on the left provides quick access to essential options such as About Us, Contact Info, Event Info, Feedback, and Admin Login each represented with user-friendly icons. A “RECENTER” button ensures users can return to their current location on the map with a single tap. On the right side of the screen, the layout features a categorized list of “B Blocks” (A, B, C, D, W) for fast reference, alongside visual markers for key facilities like the Masjid, Bus Stand, Cafeteria, and Library. At the bottom, a location search bar and a “Generate QR Code” button allow users to search for specific places and create QR codes for sharing or saving locations. A voice input button further enhances usability by enabling hands-free interaction. The interface is intuitive, visually balanced, and tailored for both new visitors and regular users of the campus.

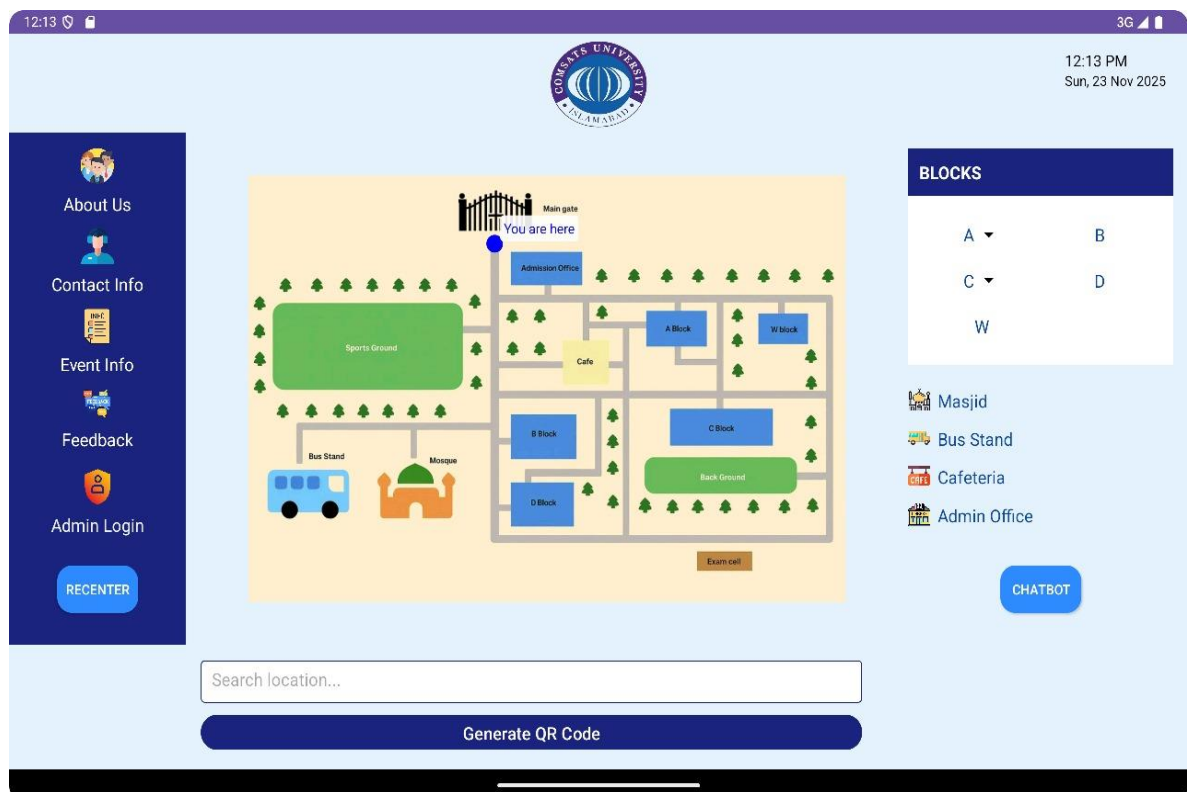


Figure 5.2 Main Screen

5.3.2.1. Main Screen-Indoor Mapping

Figure 5.3 represents the indoor mapping interface of the CUI Virtual Guide application for COMSATS University Islamabad, Sahiwal Campus. The interface shows a detailed layout of the campus ground floor, highlighting key locations such as classrooms (e.g., A3), the library, IT office, admin office, medical center, kitchen, and director's office. The current location of the user is indicated with a marker labeled "You are here," enabling easy orientation within the campus. Navigation options allow users to select different blocks (Block A and Block C) and floor levels, providing multi-level indoor navigation. The interface also includes a search bar for locating specific rooms or offices, a side menu for About Us, Contact Info, Event Info, Feedback, and Admin Login, and a "Generate QR Code" button to share the map view. Overall, this design facilitates precise indoor navigation, enhances user accessibility, and supports real-time location-based guidance.



Figure 5.3 Main Screen-Indoor Mapping

5.3.2.2. Search Destination

Figure 5.4 screenshot shows the destination selection interface of the CUI Virtual Guide app. It displays a modal dialog titled "Select destination," listing various campus locations that a user can choose to navigate to, including offices, classrooms, labs, medical center, library, and

washrooms. This interface allows users to quickly pick their desired location, which the app will then use to generate a navigation route or a QR code for easy access. The background still shows the main campus map interface, while the modal focuses attention on the selection task.

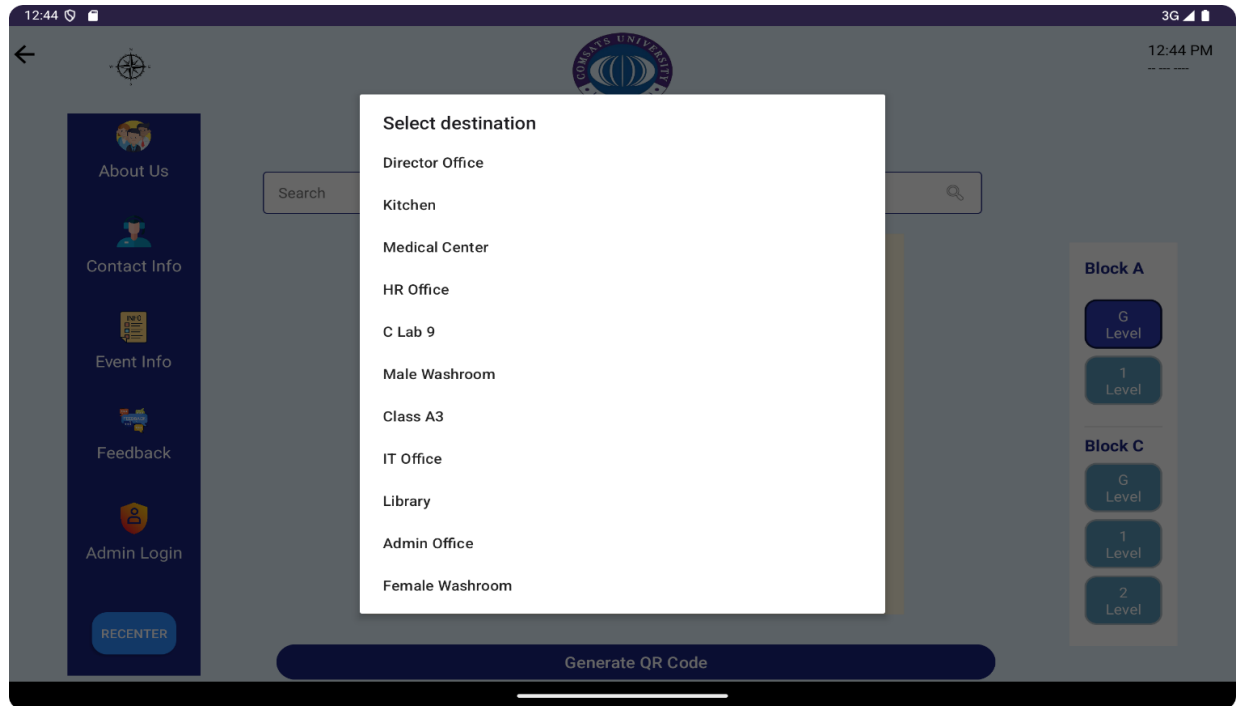


Figure 5.4 Search Destination

5.4.1. About Us Screen

Figure 5.5 screen provides an informative overview of COMSATS University Islamabad, Sahiwal Campus. It highlights the university's history, academic excellence, and strategic location between Lahore and Multan. Users can learn about the campus's establishment, its modern infrastructure, and its diverse academic programs in fields like Computer Science, Engineering, Business Administration, and Biosciences. The content reflects the institution's commitment to innovation, research, and holistic student development, offering a clear and inspiring introduction for prospective students and visitors.

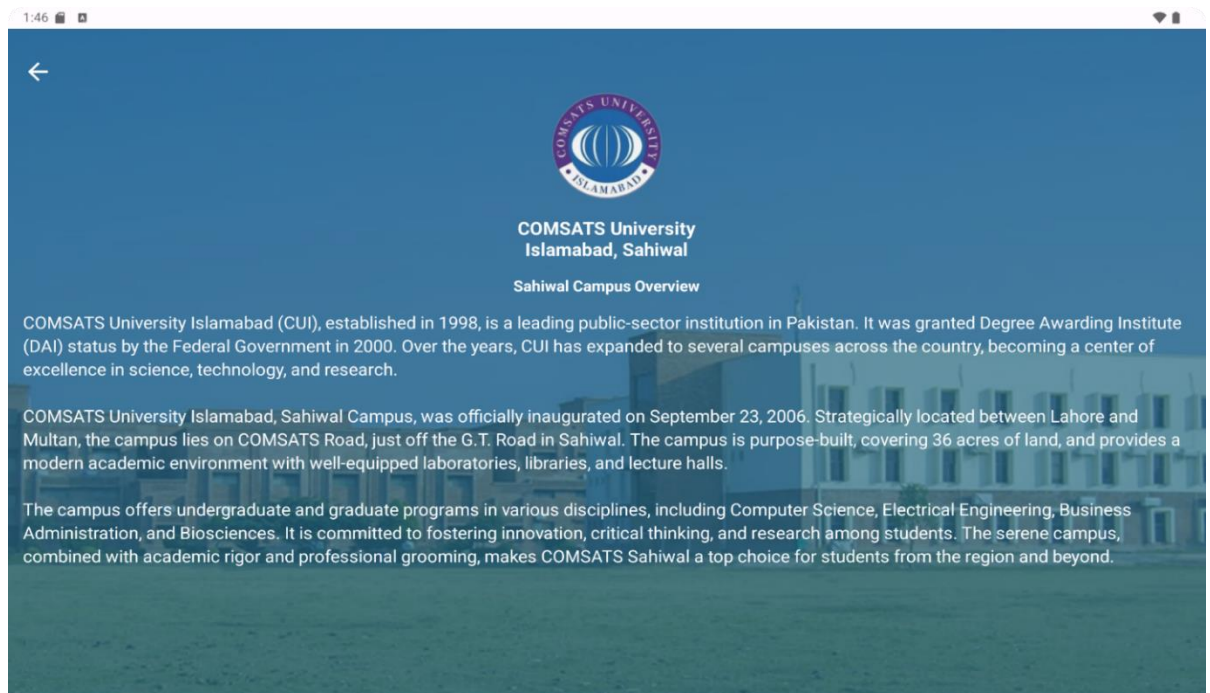


Figure 5.5 About us Screen

5.4.2. Contact Us Screen

Figure 5.6 shows a “Contact Us” screen from a mobile app featuring the COMSATS Sahiwal campus in the background. The UI includes input fields for Name, Email, Phone, and Message, overlaid on a semi-transparent dark box for readability. A prominent blue “SEND” button allows users to submit their queries. The layout uses a full-screen background image, with the university logo and title placed at the top. The design is clean, professional, and mobile-friendly.

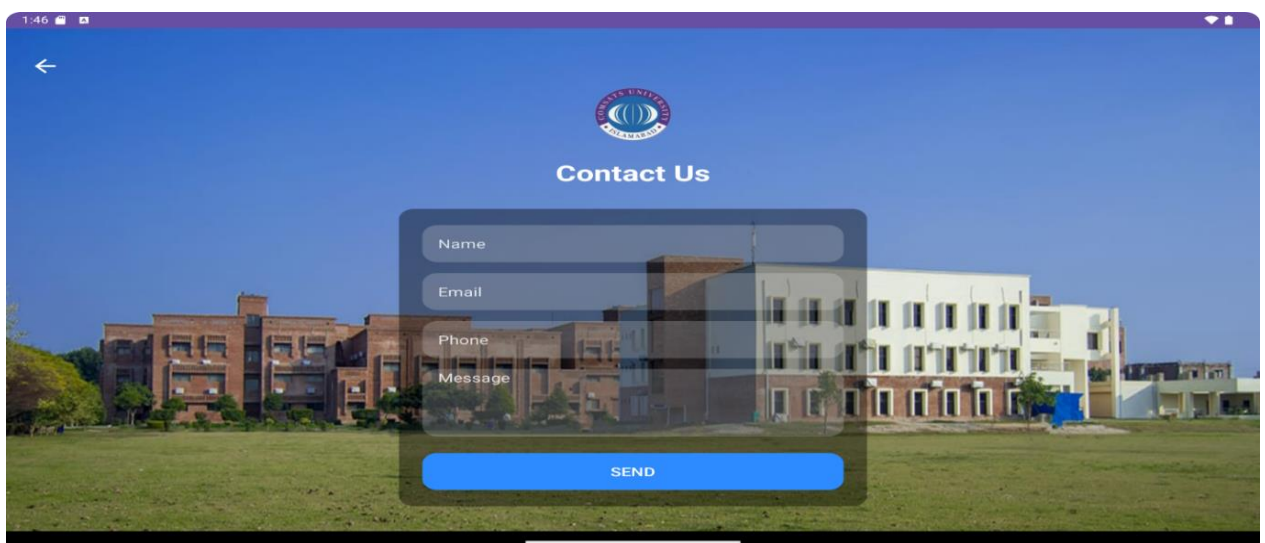


Figure 5.6 Contact us Screen

5.4.3. Event Information Screen

Figure 5.7 shows a “Event Information” screen of a mobile app for COMSATS University Islamabad, Sahiwal Campus. It features a calendar for May 2025, allowing users to view upcoming events. Below the calendar are filter buttons: All, Seminars, Sports, and Cultural, helping users sort events by type. At the bottom, Sync and Chat buttons provide options for updating events and user interaction. The background shows the university campus, maintaining a consistent visual theme.

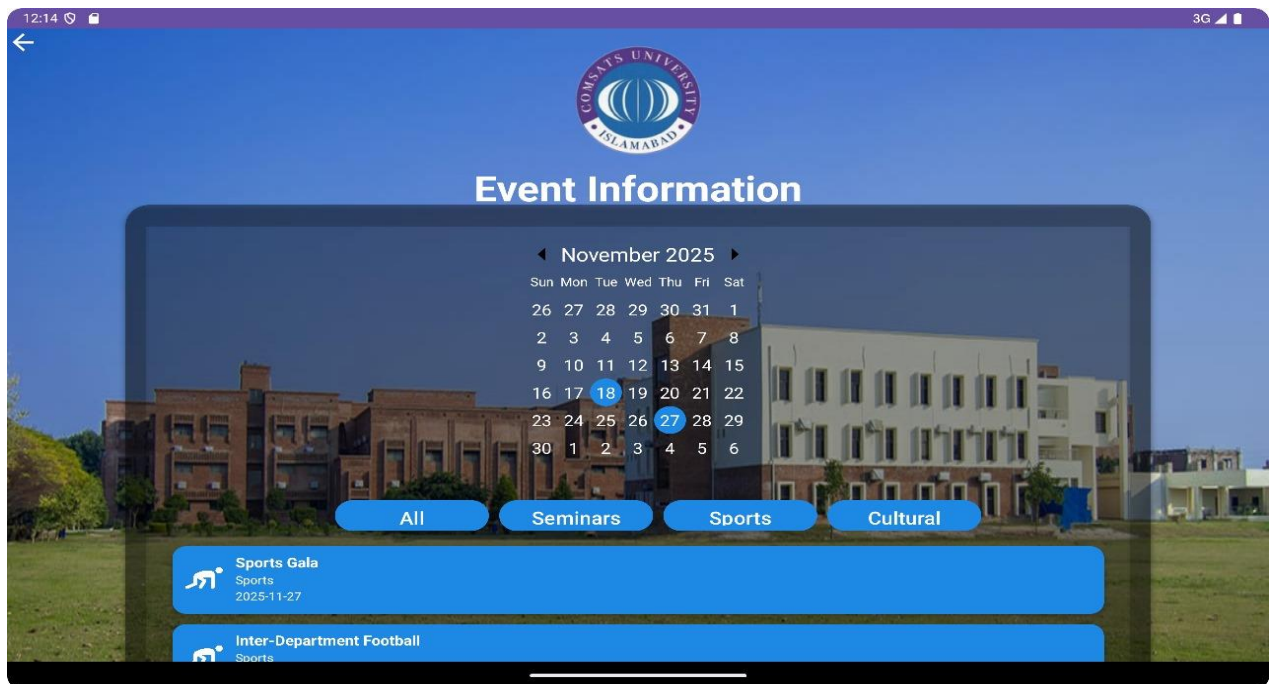


Figure 5.7 Event Information Screen

5.4.4. Feedback Screen

Figure 5.8 shows a “Feedback” screen of the COMSATS University Sahiwal mobile app. It features input fields for Full Name, Email, Suggestion, and Message, allowing users to share their opinions. Below the form is a star-based rating system for user satisfaction. The blue “SUBMIT” button lets users send their feedback. The background showcases the university campus, maintaining a clean and branded user interface.

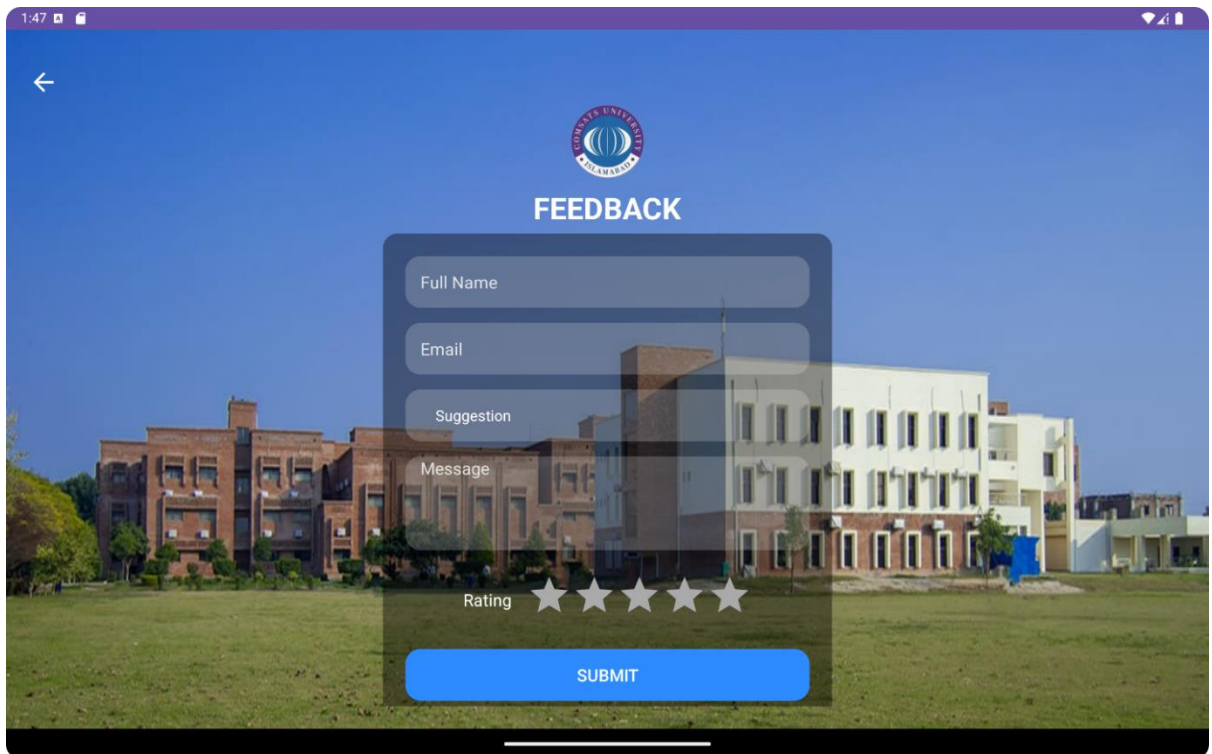


Figure 5.8 Feedback Screen

5.4.5. Admin Login Screen

Figure 5.9 shows a screen is the Admin Panel login interface of a mobile application, likely developed for COMSATS University Islamabad. It features a background image of the university campus, with a transparent login box overlay. The login form includes fields for Admin ID and Password, along with a "LOG IN" button and a "Forgot password?" link. The COMSATS logo is displayed at the top, and the design follows a clean, modern aesthetic.

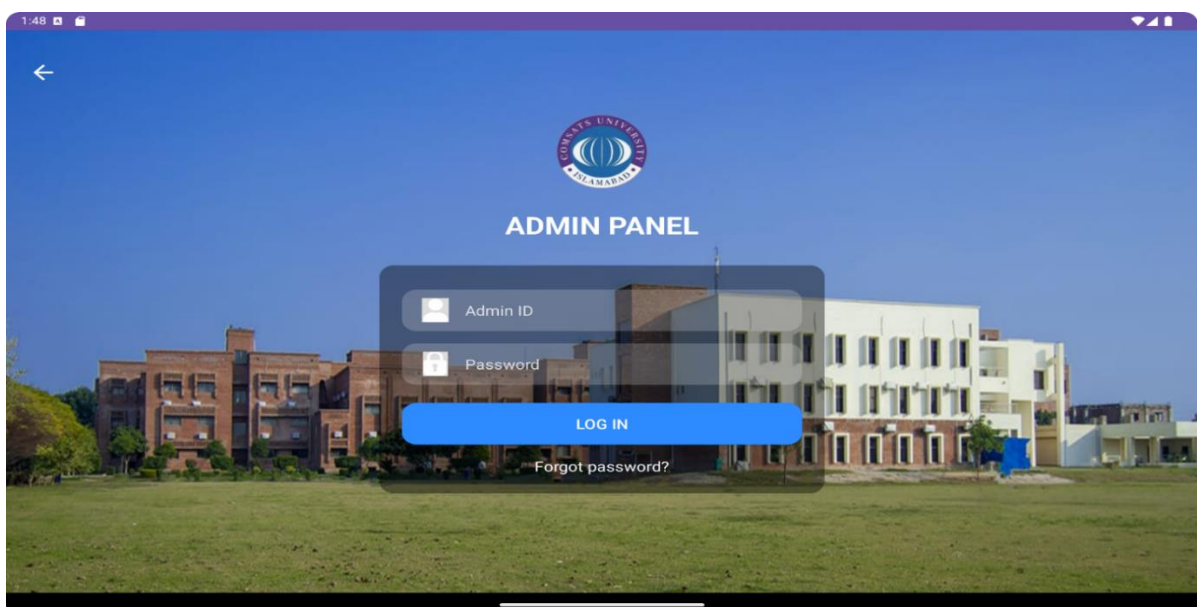


Figure 5.9 Admin Login Screen

5.4.5.1. Admin Dashboard

Figure 5.10 represents a screen represents the Admin Dashboard of the COMSATS University Virtual Guide application, displayed immediately after a successful login and designed with the same university-themed background and transparent overlay for visual consistency. The dashboard acts as the central control panel for all administrative operations, providing four main options Manage Events, View Feedback, Edit Map, and View QR Codes each represented with clear icons and buttons for easy navigation. Through these features, admins can efficiently update campus events, review user feedback submitted through the kiosk, modify or adjust map routes to keep navigation accurate, and access or regenerate QR codes used for sharing map screenshots. Overall, the Admin Dashboard ensures smooth, organized, and efficient system management while maintaining a clean and professional user interface.

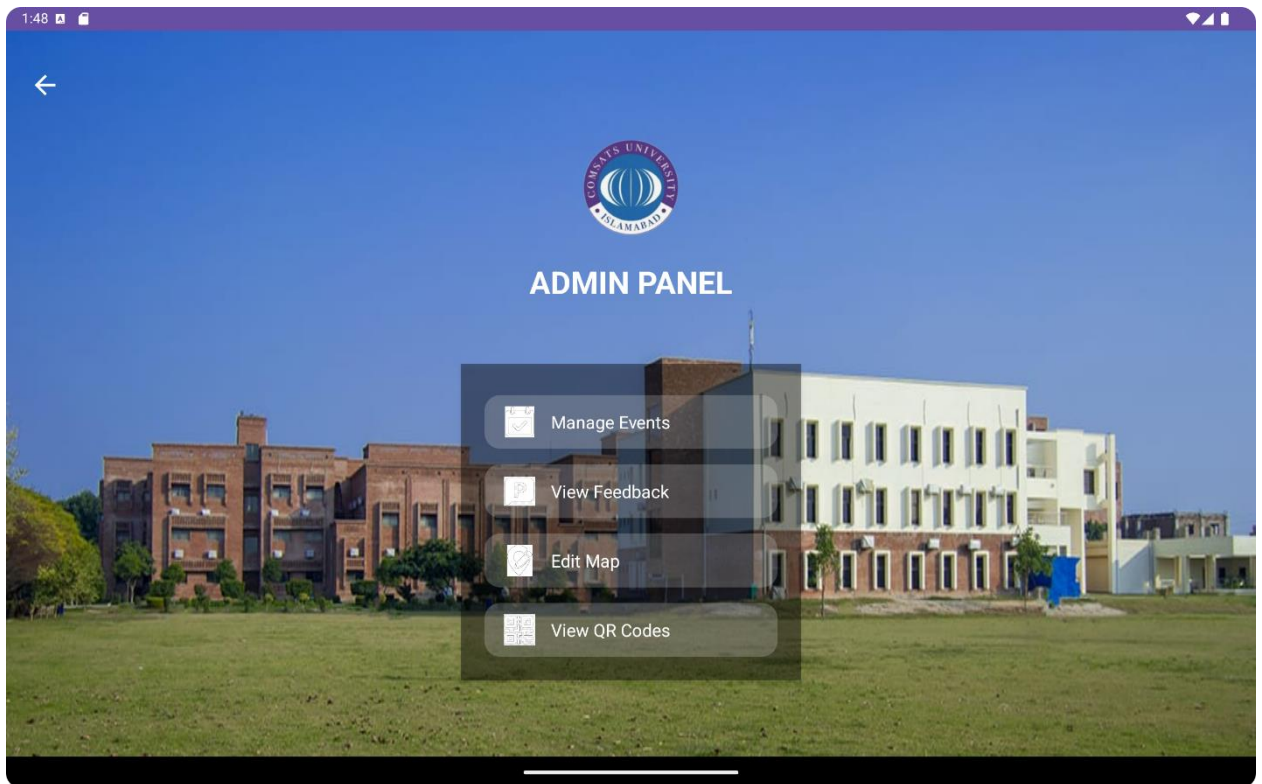


Figure 5.10 Admin Dashboard

5.4.5.2. Manage Events Screen

Figure 5.11 shows the Admin Event Management page where the admin can view all events in a list and perform actions. Each event card displays the event's name, category and provides Update and Delete options. The all events are added through real time web-scraping.

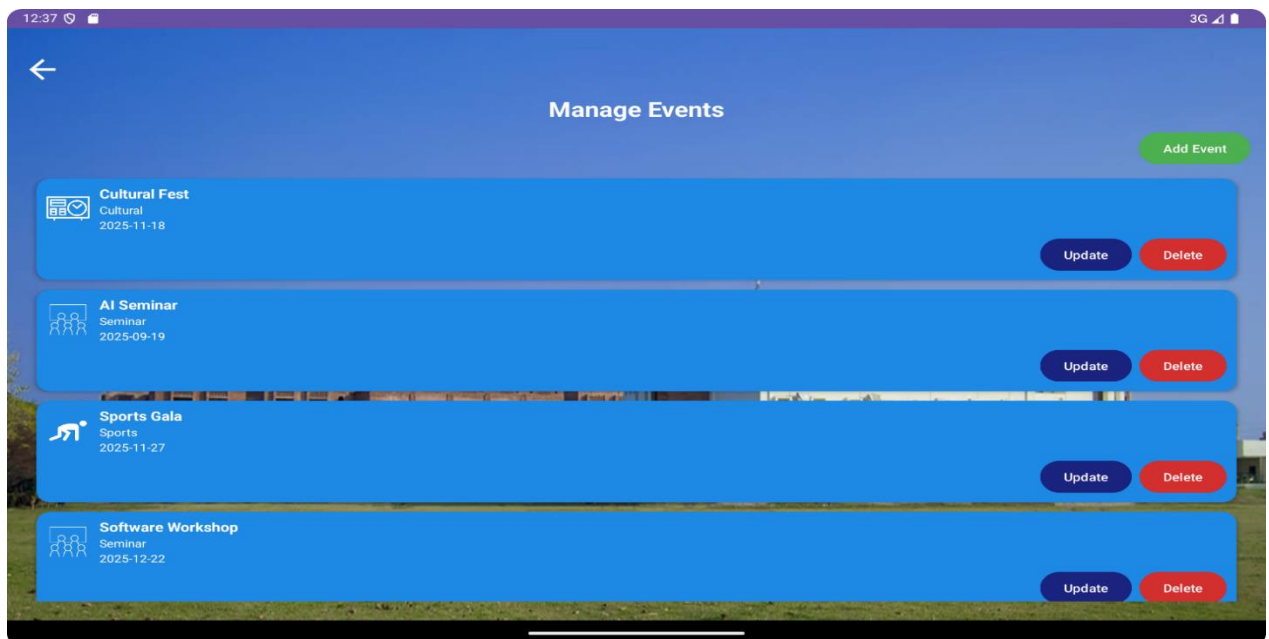


Figure 5.11 Manage Events

5.4.5.3. Manage Feedback Screen

Figure 5.12 displays the Admin Feedback Management page where the admin can view user feedback including name, email, type, message, and rating. Each feedback entry provides Reply and Delete options, allowing the admin to respond or remove feedback. The background features a campus building, giving a university-themed interface.

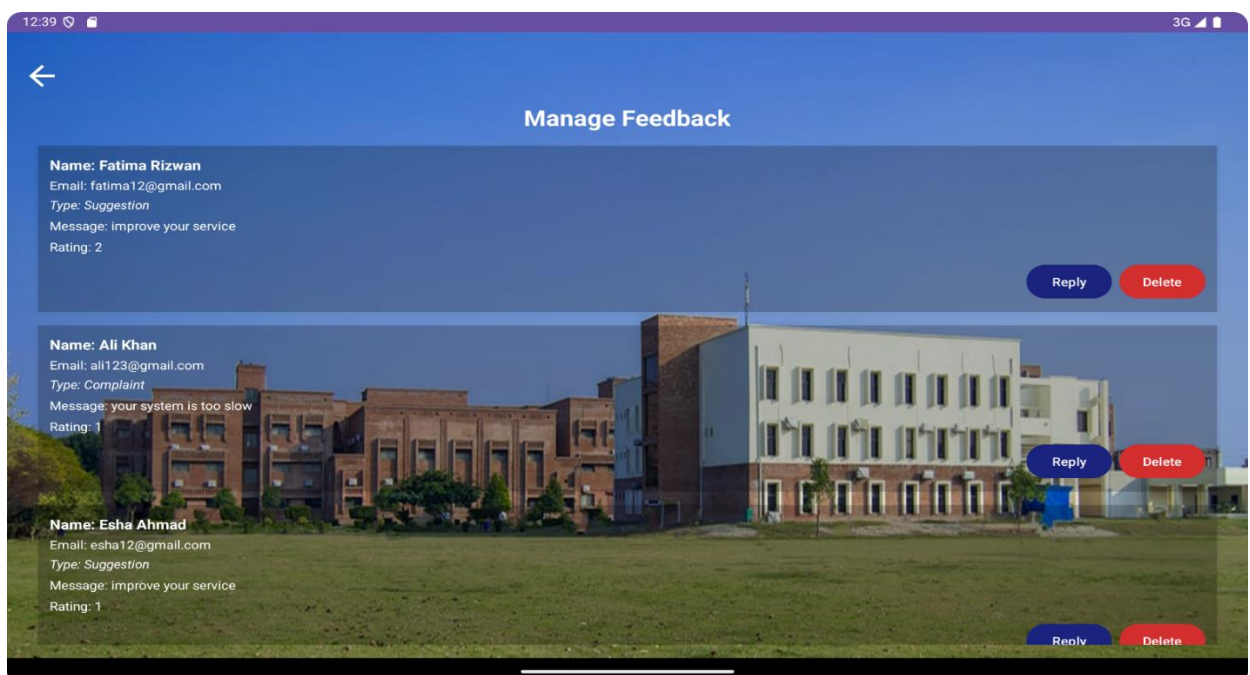


Figure 5.12 Manage Feedback

5.4.5.4. Manage QR Screen

Figure 5.13 displays the QR Code Management page where the admin can view uploaded QR code links along with their upload date and time. Each QR entry appears in a separate

card and includes a Delete button for removing unwanted QR codes. The background shows a campus building, keeping the interface visually consistent with the app theme.

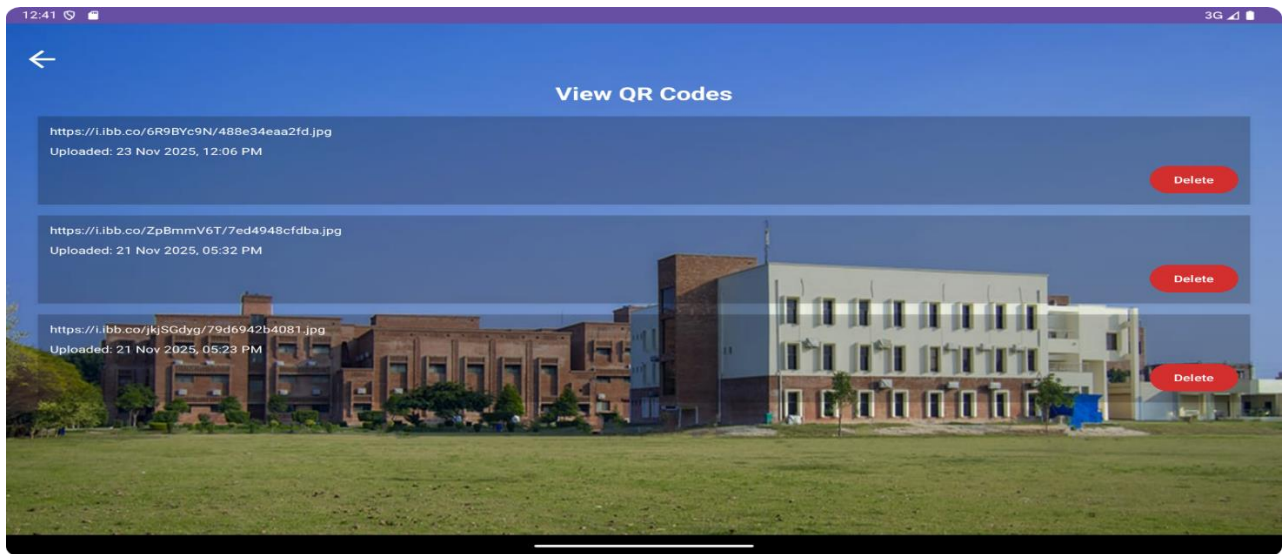


Figure 5.13 Manage QR Code

5.4.5.5. Chatbot Interface

Figure 5.14 displays the Chatbot interface, where users can interact with the virtual assistant to ask questions, receive guidance, and access campus-related information instantly. The chatbot window appears in the center of the screen, showing the conversation history along with a text input bar for typing queries. The interface is designed to provide quick responses about campus navigation, department information, events, and general queries. The layout remains visually aligned with the app's overall theme, ensuring a smooth and user-friendly experience while communicating with the virtual guide.



Figure 5.14 Chatbot Interface

5.4.5.6. QR Generate -Screen

Figure 5.15 represents the COMSATS University Islamabad (CUI) Virtual Guide app, displaying a map of the campus and a prominent QR code. This system allows users to generate and scan the QR code to transfer a pre-calculated route or navigation path directly to their personal phone for convenient, on-the-go guidance across the campus.



Figure 5.15 QR Generating - Interface

Chapter 6

Testing and Evaluation

6. Testing and Evaluation

Testing and evaluation were essential phases of the COMSATS University Virtual Guide project to ensure that the kiosk application, navigation system, admin dashboard, voice assistant, and feedback modules worked smoothly and met user expectations. The purpose of testing was to detect and resolve errors early, validate system behavior under different conditions, and confirm that all features aligned with the project requirements. A combination of manual, system, unit, functional, integration, and automated testing was performed to evaluate the performance, usability, and reliability of the system. Since the project included map navigation, Firebase connectivity, QR code generation, and TTS-based voice interactions, rigorous testing was required to ensure correct route display, fast data storage, proper voice output, and accurate admin functionalities.

6.1. Manual Testing

Manual testing was conducted to verify that each feature of the kiosk and admin system worked as intended without using any automation tools. This approach helped identify usability issues, navigation flow problems, and input handling errors that automated tools may overlook.

Objectives of Manual Testing

- To validate the correctness of individual features such as navigation, voice assistant, event display, and feedback submission.
- To identify UI/UX problems or design inconsistencies on the kiosk.
- To ensure proper error messages for invalid inputs.
- To test system behaviour with different user actions and edge cases.

Manual Testing Process in This Project

Manual testing covered key modules of the Virtual Guide System, including destination selection, route display on the map, QR code generation, voice assistant responses, feedback submission forms, and admin dashboard operations.

For example:

- Destination selection was tested by choosing different campus locations and checking whether the correct predefined route appeared on the map.
- Voice assistant commands were tested to ensure the TTS engine produced clear, accurate guidance.

- Feedback/contact forms were filled multiple times with valid and invalid inputs to verify Firebase storage and validation messages.
- Admin dashboard functions, such as viewing feedback or viewing QR codes, were manually tested to ensure smooth navigation and data retrieval.
- Manual testing helped refine the interface and ensured that both kiosk users and administrators had a seamless experience.

6.2. System testing

System testing validated the entire Virtual Guide application as a complete, integrated system. This testing examined how different modules navigation, TTS, QR generation, Firebase forms, and admin dashboard work together from end to end.

Purpose of System Testing

- To ensure smooth interaction between all system components.
- To verify end-to-end navigation workflows from user input to map output.
- To confirm that Firebase integration (feedback, contact queries, events) works reliably.
- To ensure the system meets both functional and non-functional requirements.

Key Functionalities Tested

System testing focused on core features of the Virtual Guide:

- **Map Navigation:** Displaying predefined routes correctly when a destination is selected.
- **AI Voice Assistant:** Producing accurate spoken responses via TTS.
- **QR Code Generation:** Creating QR codes containing map screenshots or URLs.
- **Feedback Submission:** Sending user feedback directly to Firebase without login.
- **Contact Form Submission:** Storing user messages in Firebase in real time.
- **Admin Dashboard:** Viewing feedback, editing map data, and accessing QR codes.

This phase also included usability testing to ensure the kiosk UI was simple and intuitive, especially for first-time visitors.

6.3. Unit Testing

Unit testing validated individual functions and modules of the system to ensure each worked correctly on its own. Table 6.1 represents a testing identified early bugs in isolated components like the TTS engine, navigation function, Firebase write operations, and QR code generator.

Unit Testing 1: Destination Navigation

Table 6.1 ensure that navigate to destination correctly.

Table 6.1 Destination Navigation

No.	Test Case	Input	Expected Result	Result
1	Select a valid destination	“CS Department”	Correct route polyline displayed on the map	Pass
2	Select an invalid or missing destination	“XYZ Hall”	Error message: “Route not found”	Pass

Unit Testing 2: Feedback Submission

Table 6.2 testing ensure that feedback is stored correctly in Firebase.

Table 6.2 Feedback Submission

No.	Test Case	Input	Expected Result	Result
1	Submit valid feedback	Name, email, message	Data stored in Firebase + success message	Pass
2	Submit empty fields	Blank message	Validation error displayed	Pass

6.4. Functional Testing

Functional testing validated that each module of the Virtual Guide performed according to the defined requirements. Each feature was tested using various valid and invalid inputs to evaluate how the system responded.

Functional Testing Focus Areas

- Destination selection and route display.
- TTS playback for greetings, building information, and directions.
- QR code creation for map screenshots.
- Firebase submissions (feedback/contact).
- Admin dashboard functions.

Functional Testing Example: Admin Login

Table 6.3 represents the admin system is role-based, functional testing ensured only authorized users could access admin tools.

Table 6.3 Admin Login

No.	Test Case	Input	Expected Result	Result
1	Login as admin	Correct email + password	Access admin dashboard	Pass
2	Login with incorrect password	Wrong password	Show “Invalid credentials”	Pass

6.5. Integration Testing

Table 6.4 represents a Integration testing verified how different modules interact and whether data flows correctly between them. Since the system includes multiple connected components Firebase, navigation maps, QR codes, and TTS testing integration points was critical.

Objectives of Integration Testing

- To check communication between UI, map, Firebase, and APIs.
- To ensure modules produce accurate outputs when used together.
- To validate cross-feature workflows such as screenshot → upload → QR code.

Table 6.4 Integration Testing

No.	Test Scenario	Expected Result	Result
1	Select destination → Generate route → Voice assistant announces destination	Route shown + TTS output plays	Pass
2	Take map screenshot → Upload via ImgBB → Generate QR	QR displays correctly	Pass
3	Submit feedback → Admin views it in dashboard	Data appears instantly	Pass

6.6. Automated Testing

Table 6.5 represents a automated testing was performed on selected functionalities such as navigation buttons, admin login, and UI navigation flow to ensure consistency. A simple automation tool (like Espresso or UI Automator in Android Studio) was used to simulate user interactions with the kiosk interface.

Why Automated Testing?

- Speeds up repeated tests like navigation and login.
- Detects UI navigation issues early.
- Ensures consistent behaviour across multiple runs.

- Reduces manual testing workload.

Automated Test Coverage

Automated scripts tested:

- Destination selection buttons.
- Basic navigation between screens.
- Admin login.
- TTS trigger button interactions.

Table 6.5 Automated Testing

Tool Name	Tool Description	Applied On	Results
UI Automator	Android UI testing tool	Admin login, screen navigation, destination buttons	All scripts executed successfully; tests passed

Automated testing confirmed the stability and consistency of essential kiosk interactions.

Chapter 7

Conclusion And Future Work

7. Conclusion

In conclusion, the CUI Virtual Guide was successfully developed as a user-friendly platform aimed at providing students, visitors, and staff with seamless campus navigation and information access. The application caters to a diverse audience by offering intuitive navigation between buildings, real-time directions, AI-assisted guidance, QR-based map sharing, and instant access to campus information. Key functionalities were implemented to improve user satisfaction, including route visualization on the interactive campus map, AI voice assistant interactions, QR code generation for sharing locations, and feedback/contact submission to allow user suggestions and inquiries.

The development and deployment of the CUI Virtual Guide mark a significant milestone in creating a modern, accessible, and interactive campus navigation system. The primary objective was to provide a solution for users who do not know the location of classrooms, departments, or facilities on campus, and this objective has been achieved through a carefully designed, tested, and user-friendly interface.

User-Friendly and Accessible Design

The application was designed with a focus on accessibility and usability. The interface includes an intuitive layout for selecting destinations, viewing optimal routes, and interacting with the AI voice assistant. QR code functionality allows users to capture and share location information, while feedback and contact forms ensure communication with administrators. Accessibility considerations, such as readable fonts, high-contrast visual cues, and clear voice guidance, make the application usable by individuals with differing abilities.

Comprehensive Feature Set for Customer Satisfaction

Key functionalities that enhance the user experience were successfully integrated into the platform:

AI Voice Assistant: Provides audible guidance and interactive responses to user queries.

Navigation System: Generates optimal routes between campus locations using predefined polylines.

QR Code Generation: Allows users to share map screenshots via QR codes for mobile access.

Feedback and Contact Forms: Enables users to submit suggestions and queries, which are stored securely in Firebase for administrative review.

Backend Administration and Content Management

The admin interface provides control over campus information management, including adding or updating locations, managing QR codes, and reviewing feedback submissions. This ensures that the application remains accurate, up-to-date, and relevant for users.

Meeting Primary Objectives

Through these features and extensive testing including manual and automated methods the CUI Virtual Guide has met its primary objectives of providing efficient campus navigation, real-time guidance, and accessible information for visitors, students, and staff.

7.1. Future Work

Several enhancements can be considered to elevate the capabilities, usability, and overall effectiveness of the CUI Virtual Guide, making it a more intelligent, user-centric, and scalable campus navigation system.

Real-Time Location Tracking is one of the most impactful improvements that could be implemented. Currently, users rely on static or predefined routes to navigate the campus. By integrating live location detection, users would be able to see their exact position on the campus map in real-time, significantly improving navigation accuracy. This could be achieved through technologies such as GPS for outdoor areas and indoor positioning systems, including Bluetooth beacons, Wi-Fi triangulation, or RFID, for precise indoor navigation. Real-time tracking would allow the guide to dynamically update the user's location as they move, enabling step-by-step navigation and reducing confusion or misdirection, especially for new students or visitors unfamiliar with the campus layout.

Multilingual Voice Support represents another key enhancement that would improve accessibility for a broader audience. By incorporating multiple languages into the AI voice assistant, the system would cater not only to local students but also to international visitors and faculty. This would involve integrating additional language datasets and speech synthesis models capable of generating natural, context-aware voice outputs. Multilingual support would ensure that voice-guided navigation, directions to campus facilities, and responses to general queries are understandable to users from diverse linguistic backgrounds, enhancing inclusivity and user satisfaction.

Mobile App Integration would allow users to access the Virtual Guide on the go, independent of kiosk terminals. A dedicated mobile application for iOS and Android

could provide additional functionalities such as push notifications for campus events, personalized alerts based on user location, and offline map access for areas with weak internet connectivity. Users could plan their campus routes ahead of time or receive real-time guidance while walking around the campus, making the system more flexible and convenient. Mobile integration would also allow continuous updates and remote access to campus resources, further increasing engagement and utility.

Advanced Route Optimization would improve the intelligence of the navigation system. While the current version relies on predefined routes, future implementations could incorporate dynamic pathfinding algorithms, such as Dijkstra's or A*, to calculate the shortest, fastest, or most accessible route between any two locations. Such algorithms could consider factors like temporary obstructions, ongoing construction, or accessibility requirements for individuals with mobility challenges. By dynamically recalculating optimal paths, the system could provide more reliable and efficient navigation, enhancing user trust and experience.

Enhanced QR Functionality is another promising feature for expanding the interactivity of the system. Currently, QR codes allow users to access map screenshots, but future versions could link QR codes to additional campus resources, such as event schedules, departmental webpages, or facility usage information. By scanning a QR code at different locations, users could instantly retrieve relevant content, such as room availability, upcoming lectures, or announcements, turning the campus guide into a fully interactive, multi-purpose information system.

Data Analytics and Reporting could empower administrators to make data-driven decisions. By analyzing user interactions, feedback submissions, frequently visited locations, and common navigation requests, administrators could identify trends and patterns in campus usage. This information could be used to optimize signage, improve campus infrastructure, or update routing priorities. Additionally, dashboards with real-time reporting capabilities could help in assessing the effectiveness of system updates and prioritizing features based on user behavior, making the Virtual Guide not only user-centric but also management-friendly.

Accessibility Enhancements are essential for creating an inclusive campus navigation system. Future improvements could include more robust text-to-speech functionality for

visually impaired users, customizable voice instructions, high-contrast interface themes for better readability, and adjustable font sizes. Incorporating these features would ensure that all students, staff, and visitors, regardless of physical ability or visual limitations, can effectively use the system. Such accessibility-focused updates would not only comply with modern usability standards but also foster an equitable campus environment.

In summary, these proposed enhancements ranging from real-time location tracking and multilingual support to advanced route optimization, enhanced QR interactivity, data-driven analytics, and improved accessibility would transform the CUI Virtual Guide into a highly intelligent, interactive, and scalable solution. By implementing these features, the system would provide a seamless, engaging, and fully supportive experience for all campus users, ensuring that it meets both current needs and future demands in campus navigation and user assistance. considered to elevate the capabilities, usability, and overall effectiveness of the CUI Virtual Guide, making it a more intelligent, user-centric, and scalable campus navigation system.

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Chapter 8

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